

Adverse Selection in (Un)Subsidised Health Insurance: Evidence from Nepal's Age-70 Threshold*

Sabin Subedi [†]

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A growing number of governments in low- and middle-income countries are building universal health coverage on contributory health insurance schemes. The mandate behind them is rarely enforced, so enrolment is voluntary in practice and enrollees sort on the premium. How strongly that price sorts them on risk determines what the pool costs to insure. This paper provides the first quasi-experimental estimate of adverse selection in a national scheme in a low- or middle-income country, using a statutory premium waiver at age 70 in Nepal's National Health Insurance Programme in a regression discontinuity in age at enrolment. Coverage nearly doubles at the threshold, while the waiver extends it into places that are poorer and farther from care. Cost per enrollee falls across every claims margin. Because enrolment itself responds to the price, these compare pools of differing composition rather than a fixed population's use of care. The marginal cost of the enrollees the zero price draws in is significantly below the average cost of the paying pool, which is adverse selection into the paid programme. The pool was adversely selected while both remedies proposed against selection in these markets, household bundling and restricted enrolment windows, were in force. These findings show that selection in these markets runs through the price, so remedies that leave the price alone cannot remove it.

Keywords: Adverse Selection, Health Insurance, Regression Discontinuity, Nepal

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*Correspondence: sabin.subedi@strath.ac.uk. All mistakes are my own.

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[†] : Affiliated to University of Strathclyde

1 – Introduction

Governments in low- and middle-income countries have expanded public health financing sharply over the past two decades as part of a broader effort to build universal health coverage, and most have built that expansion around national insurance rather than the health budget alone (Lagomarsino et al. 2012; Kurowski et al. 2023). The appeal of insurance is that premiums collected before treatment would convert the payments patients were already making at the point of care into public revenue for the health system, and coverage would protect households against the impoverishment that a health shock can bring (Das and Do 2023; Chetty 2009). Since most of the population sits outside the tax net, or the payroll from which a contribution could be withheld, the schemes are contributory and compulsory in law, but they operate as voluntary schemes in practice (Banerjee et al. 2021; International Labour Organization 2023). Ghana, Kenya, the Philippines, Vietnam and Indonesia each have a contributory scheme that the law makes compulsory and that collects premiums directly from those it enrolls, whether it enrolls them as individuals or as households (Lagomarsino et al. 2012), and Nepal’s programme, the setting of this paper, is built the same way.

Voluntary enrolment is what turns the composition of the risk pool into a policy problem. Coverage that has to be bought is bought by few, and disproportionately by those who expect to use it (Akerlof 1970; Rothschild and Stiglitz 1976), so enrolment stays low and the pool that does enrol costs more per member than the population it is drawn from. The fiscal rationale has not survived contact with that fact, because individuals have proved unwilling to pay unsubsidised premiums (Das and Do 2023), and governments have met the gap from general revenue, subsidising the poor and covering deficits as they arise. How far a scheme can extend coverage before it turns against it therefore depends on how strongly the premium sorts on risk, which makes the extent of adverse selection a question about whether these schemes can be sustained at all.¹

Evidence on adverse selection in low- and middle-income insurance markets is limited and mixed. Some studies show evidence of adverse selection, with sicker people enrolling into health

1. Nepal’s programme illustrates the bind. Arrears owed to empanelled hospitals passed NPR 11 billion by late 2025 and more than fifty hospitals had suspended acceptance of insurance cards by early 2026, while the Health Insurance Board responded by capping outpatient coverage at NPR 25,000 within the NPR 100,000 annual ceiling (The Kathmandu Post 2026, 2025).

insurance (Fischer, Frölich, and Landmann 2023; Thornton et al. 2010; Yao, Schmit, and Sydnor 2017; Asuming, Kim, and Sim 2024). Other studies report little or none (Wagstaff et al. 2016; Banerjee, Duflo, and Hornbeck 2014). Part of the disagreement reflects how challenging selection is to identify. The standard test asks whether coverage and realised risk are positively correlated (Chiappori and Salanié 2000), which establishes that selection is present without signing or sizing it. Doing either requires variation in the price of coverage that moves who enrolls while leaving their health unchanged (Einav and Finkelstein 2011), and national schemes rarely supply such variation outside an experiment. Where selection has been found, two design remedies recur: bundling coverage so that it cannot be bought for one member alone (Fischer, Frölich, and Landmann 2023), and restricting when enrolment may happen, so that coverage cannot be bought at the moment it is needed (Banerjee et al. 2021).

Nepal's National Health Insurance Programme (NHIP) carries both remedies and supplies the price variation.² The programme is contributory and voluntary in practice. A household of up to five members pays an annual premium of NPR 3,500, coverage is bundled across its members, and enrolment is confined to four fixed start dates a year and must be arranged a quarter in advance, so a household cannot wait until it needs care and then join for immediate benefit. Since 2019 the premium has been waived for anyone aged 70 or over at the start of an enrolment spell. The price of coverage falls to zero at an age the programme's own rule fixes in advance, and it falls for reasons that have nothing to do with any individual's health. If the enrollees the zero price draws in claim less than the enrollees the positive price had already retained, then the premium had been screening the low-cost people out and holding the costlier in (Einav and Finkelstein 2011).

I estimate that comparison in a local linear regression discontinuity in age at enrolment, setting spells that begin just before an enrollee's seventieth birthday against those beginning just after. I compile a dataset from the universe of the enrolment register held by the Health Insurance Board and the reimbursement claims filed against it. I geocode the enrollee's ward and the empanelled facilities in it to measure the enrollee's distance to care and the number of empanelled facilities within 10 km of their ward. To my knowledge, this is the first study to compile a dataset of this kind. Since the data cover only those who enrol, the parameter needs stating with care. It compares the pool that enrolls at a positive premium just below 70 with the pool that enrolls at a

2. Subedi (2026) provide a detailed account of the programme's design, implementation and its effects on utilisation.

zero premium just above. It is a pool-composition contrast rather than the effect of free insurance on a fixed population's use of care. The enrolled population itself jumps at the threshold, and that response is the object of interest. A price change in a voluntary market works by changing who buys, so the discontinuity measures what the re-sorting does to the risk the programme carries, which is what sets its cost per enrollee.

I have three main findings. First, coverage rises sharply at the threshold, from 22 percent of the age-eligible population just below 70 to 42 percent just above, so the waiver draws in a fifth of that population. Enrollees at the threshold live 0.54 km farther from their assigned first point of contact, have 0.74 fewer empanelled facilities within 10 km of their ward, and live in poorer local units. The waiver extends coverage into the places where the gap was. Second, claims per enrollee fall over the same interval. Total claims fall by NPR 921.5, a decline of 17.4 percent; the probability of any claim falls by 5.9 percentage points, a decline of 11.8 percent; and the mean claim falls by NPR 161.6, a decline of 15.0 percent. Inpatient stays, which an enrollee cannot award herself, fall by more in proportional terms than most other margins, with the probability of inpatient stays down by 17.0 percent and the number of inpatient claims down by 19.4 percent. I show that these results are robust to a range of bandwidths, donut estimations, polynomial orders and kernel weights, and they are absent at placebo thresholds.

Third, I estimate the marginal cost of the enrollees the waiver draws in by adapting the framework of Einav and Finkelstein (2011) to the discontinuity design. Under the assumptions set out in Section 5, I find that the marginal cost of the enrollees the waiver draws in is NPR 3,358 each in claims against NPR 5,269 for the pool that was already paying, a ratio of 0.637. The closest quasi-experimental comparison is a premium discontinuity in the Massachusetts subsidised exchange, where the same ratio is 0.59.³

This paper makes three contributions. First, it provides what is to my knowledge the first quasi-experimental estimate of adverse selection in a national health insurance scheme in a low- or middle-income country, using the age-70 premium waiver and individual-level registration and claims records, and contributes to the literature that identifies selection from the cost curve (Einav, Finkelstein, and Cullen 2010; Hackmann, Kolstad, and Kowalski 2015; Panhans 2019; Finkelstein, Hendren, and Shepard 2019), whose estimates are drawn from high-income settings. The exception

3. Constructed from the estimates of Finkelstein, Hendren, and Shepard (2019) rather than reported by them.

is Fischer, Frölich, and Landmann (2023), who identify adverse selection in a low-income insurance market in Pakistan, but from a randomised experiment on a micro-insurer rather than from a price change administered across a national scheme. Second, it contributes to the literature on take-up of subsidised health insurance in low- and middle-income countries (Thornton et al. 2010; Wagstaff et al. 2016; Banerjee et al. 2021), where the evidence is almost entirely experimental, by estimating the enrolment response to a statutory price cut in a national scheme. It contributes to work on who public insurance reaches (Miller, Pinto, and Vera-Hernández 2013; Dupas and Jain 2024) by using individual-level registration records to characterise the enrollees the zero price draws in, in their access to care and in the poverty of the places they live. Third, it contributes to the literature on design-based defences against selection (Fischer, Frölich, and Landmann 2023; Banerjee et al. 2021) by testing them where they bind. NHIP bundles coverage at the household level and confines enrolment to fixed windows, both in force below 70, and the pool insured under them is adversely selected even so (Section 5).

1.1. Related Literature

Selection in insurance markets has been measured in two ways. The first-generation test is the positive correlation between coverage and realised risk (Chiappori and Salanié 2000), which establishes that selection is present but neither signs nor sizes it, and which can return a null when selection on preferences offsets selection on risk (Finkelstein and McGarry 2006). I follow the approach that reads selection off the cost curve instead: where the enrollees a price cut draws in are cheaper than the pool already insured, average cost falls as coverage expands and the market is adversely selected (Einav and Finkelstein 2011). Implementations have used variation in employer premiums (Einav, Finkelstein, and Cullen 2010), the introduction of a mandate (Hackmann, Kolstad, and Kowalski 2015), and a discontinuity in premiums across rating-area boundaries (Panhans 2019). Evidence from low- and middle-income settings is thinner, comes mainly from micro-insurers and field experiments, and does not point one way. Enrolment timed to planned pregnancy drove losses at a Pakistani micro-insurer (Yao, Schmit, and Sydnor 2017); in the same country, an experiment found adverse selection where policies were sold to individuals, and household bundling almost eliminated it (Fischer, Frölich, and Landmann 2023); and a Nicaraguan scheme for informal

workers drew enrollees with higher expected use (Thornton et al. 2010). A Ghanaian subsidy experiment drew in people who were ex-ante more ill, though its authors attribute the long-run utilisation difference to behavioural responses rather than to selective enrolment (Asuming, Kim, and Sim 2024). Against these, a randomised encouragement in Vietnam moved enrolment by about a percentage point, with only the previously sick responding, and its authors warn that such interventions might exacerbate selection (Wagstaff et al. 2016), and an Indian experiment that bundled insurance with microfinance found demand so weak that clients left microfinance rather than pay for cover, leaving no selection to measure (Banerjee, Duflo, and Hornbeck 2014).

A second literature asks what national schemes achieve as they expand, and the price of enrolment runs through it. Two decades of insurance expansion in low- and middle-income countries have raised utilisation and lowered out-of-pocket spending without reliably improving health, in part because providers respond to the money as much as patients respond to the price (Das and Do 2023), a pattern the randomised evaluation of India's national scheme also returns (Malani et al. 2021). Take-up is the recurring constraint, and Indonesia is the closest case to this study. Banerjee et al. (2021) randomise premium subsidies and registration assistance under a contributory scheme with a weakly enforced mandate. Left to themselves, the households that enrolled were disproportionately those already chronically sick or needing immediate treatment, and many bought a few months of coverage around a health emergency and then let it lapse, driving cost per enrollee to levels the scheme could not sustain. Time-limited subsidies raised enrolment and drew in enrollees of lower expected cost, in part by disrupting that timing. Registration assistance also raised enrolment, but raised attempted enrolment by much more, since over half of the households that tried to enrol did not manage it, so administrative capacity binds alongside price. Their subsidy is temporary and experimental and mine is permanent and statutory, which is the sense in which the result here tests whether what they find is a property of one design or of pricing a voluntary scheme at all.

The design has a direct precedent in age-eligibility thresholds in health insurance. Card, Dobkin, and Maestas (2008, 2009) use the same design at 65 in the United States. What they recover is a utilisation response on a fixed population, whereas this study is a composition contrast conditional on enrolment. The design has been used on a national scheme in a middle-income country before: Miller, Pinto, and Vera-Hernández (2013) exploit the eligibility threshold of

Colombia's Régimen Subsidiado, though what they recover is the effect of coverage on those who gain it rather than the composition of the pool that takes it up. Within my own frame the closest precedents use a discontinuity in price to trace the cost curve directly (Panhans 2019; Finkelstein, Hendren, and Shepard 2019), and it is the second of these that makes a direct comparison to the ratio of marginal to average cost I recover here.

The rest of the paper proceeds as follows. Section 2 describes the programme and the data. Section 3 sets out the design and the tests of its validity. Section 4 reports what happens at the threshold to coverage, to cost and to the health of the enrollees. Section 5 takes those estimates into the selection framework. Section 6 reports the falsification tests and the threats to identification, and Section 7 concludes.

2 — Setting and Data

2.1. Nepal and the National Health Insurance Programme

Nepal is a lower-middle-income country of 29.2 million people (National Statistics Office 2021). Gross national income per capita reached USD 1,380 in 2022 (World Bank 2025), and out-of-pocket payments account for more than half of total health spending, at USD 213 per capita (WHO 2022). The health care system combines public and private providers organised in three tiers: primary health care centres, district hospitals, and tertiary hospitals. Reducing financial exposure to health shocks is a central policy priority, and the National Health Insurance Programme (NHIP) is the government's flagship vehicle for it.

NHIP is a contributory public insurance scheme launched in 2016 and administered by the Health Insurance Board (HIB) since 2017. The programme was piloted in three districts and reached all 77 districts by 2022, covering about a third of households by the end of that year (Ayer et al. 2024). Enrolment is at the household level: a household of up to five members pays an annual premium of NPR 3,500 (USD 26), and each member beyond the fifth costs an additional NPR 700. In return the household receives cashless coverage with an annual benefit ceiling of NPR 100,000 (USD 750), shared among its members, spanning preventive, curative, inpatient, emergency, surgical, diagnostic, and rehabilitation services (HIB 2024). Several groups receive

subsidised or free coverage: female community health volunteers pay half the premium, while ultra-poor households, disabled individuals, and patients with selected terminal diseases are fully exempt (HIB 2024).

Coverage is delivered through a network of empanelled public and private facilities. A household enrolls in person and each member is issued an insurance card that identifies them to providers. At the point of care an insured member presents the card and pays nothing: throughout the study period the programme is fully cashless, with no copayment or deductible. The treating facility, not the patient, files a claim with the HIB and is reimbursed directly at pre-specified rates covering the visit, physician fees, diagnostics, medicines, and the other services in the benefit package. NHIP is a single public insurer: the HIB administers enrolment, empanelment, and reimbursement through one central information system, and it is this system that generates the administrative records this paper uses.

2.2. The Age-70 Discontinuity

Since April 2019, NHIP waives the premium for citizens aged 70 and above and assigns them an individual benefit ceiling of NPR 100,000, separate from their household's allocation. Table 1 summarises the change in insurance terms at the threshold. Two things happen at once: the price of insurance falls to zero, and the coverage *unit* changes from the household to the individual. At no age can an enrollee choose a more or less generous plan (the benefit package is uniform on both sides of the threshold), so the discontinuity is in price and coverage unit, not in any chosen quantity of insurance.

Table 1 — Insurance Terms Below and Above Age 70

	Below 70	70 and above
Annual premium	NPR 3,500 (USD 26)	Free
Benefit ceiling	NPR 100,000 per household	NPR 100,000 per individual
Coverage unit	Household	Individual

Note: Premium of NPR 3,500 applies to households of up to five members; each additional member costs NPR 700. At age 70 the individual leaves the household plan and receives a separate free individual policy.

The household premium structure makes the budgetary saving at age 70 depend on household

composition, and the dependence is not the simple per-person average. When a member of a single-person household turns 70, the household's premium obligation disappears entirely: the marginal saving is the full NPR 3,500. When a member of a two- to five-person household turns 70, the remaining members still owe NPR 3,500, so the household's marginal saving is approximately zero. Only for households above five members does removing the 70-year-old save the NPR 700 increment. The per-person price (premium divided by members) falls from NPR 3,500 to NPR 700 across the one-to-five range, but that average is not the marginal price any household saves when one member crosses the threshold. Nothing that follows requires choosing between the two. The selection analysis of Section 5 is identified from coverage and realised costs alone, so no price enters the calculation; all the design asks of the premium schedule is that the price of coverage falls to zero at 70 for everyone, whatever saving each household books.

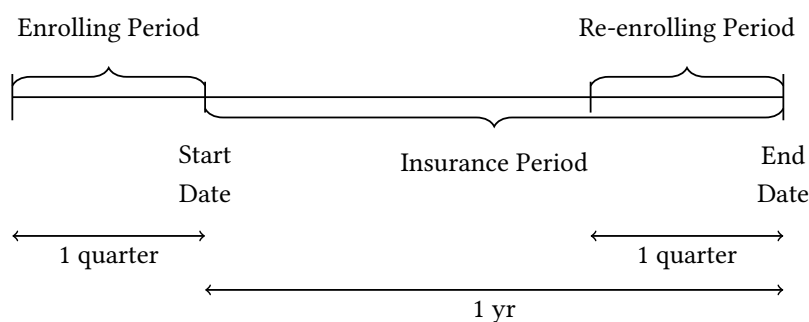
For the individual crossing the threshold, the change is unambiguous: her share of the premium obligation lapses (even where, as in two- to five-member households, the household's outlay is unchanged), and she gains an individual NPR 100,000 entitlement that no longer competes with her family's claims. Whether the price or the coverage component drives behaviour is an empirical question: benefit ceilings are slack for nearly all enrollees (annual claims reach the NPR 100,000 ceiling in fewer than 0.1 percent of spells on either side of the threshold, and the 90th percentile of claims is NPR 20,317 for household-spells below 70), so the change in the coverage unit at 70 has little room to bind. Section 5.4 takes this up as a condition on the cost comparison.

2.3. Enrolment Mechanics

NHIP operates on a quarterly enrolment cycle (Figure 1). Insurance spells start on one of four dates per year (February, May, August, or November), registration opens one quarter before each start date, and renewal is permitted only during the quarter before the current spell expires, with the new spell beginning the day after the previous one ends. The structure does more than prevent mid-spell re-enrolment. Because coverage begins only on a fixed date and must be arranged a quarter in advance, an enrollee cannot wait until she needs care and then join for immediate benefit; restricting the timing of entry in this way is one of the devices through which the programme guards against adverse selection.

Eligibility for the free premium is determined by age at the spell’s *start date*: an individual aged 70 or above when her spell begins pays nothing. Age at the start date, measured in quarters, is therefore the running variable for the research design. This timing restriction is one of the two safeguards NHIP embeds against selection, the other being household bundling. Both are in force below the age-70 threshold, and Section 5 shows that the pool insured under them is adversely selected even so.

Figure 1 – NHIP Enrolment Timeline



2.4. Data

2.4.1. Administrative data. I use administrative data from the Health Insurance Board covering the universe of NHIP enrollees and reimbursed claims from the programme’s inception through May 2025. The data consist of two linked files. The registration file records each enrolment spell: individual and household identifiers, date of birth, gender, enrolment and insurance-period dates, premium paid, registration category, and first point-of-contact facility. The claims file records each reimbursed service: claim date, amount, ICD-10 diagnosis code, hospital, and service type.

The unit of analysis is the enrolment spell. The analysis sample consists of all spells with start dates between May 2023 and May 2024, a window chosen so that the programme and its rules are uniform throughout. By 2022 NHIP had reached all 77 districts, so this cohort is drawn from a nationally complete programme rather than one still rolling out. The benefit rules do not change within the window and care is fully cashless: copayments were introduced only in 2025, after the sample ends. The claims file records reimbursements rather than services, so a claim appears only once it has been paid. Because reimbursement runs several months behind service, spells ending late in the sample are observed less completely than those ending early, which understates the

level of claims. It does not bias the discontinuity, because everyone in an enrolment cohort shares a start date and therefore the same observation window, so the incompleteness falls equally on both sides of the threshold. Because the window spans a single enrolment year, each individual enters the sample once, making the design a cross-section rather than a panel of repeated spells. For each spell I aggregate claims into spell-level outcomes (an indicator for any claim, the number of claims, the total amount claimed, and the mean claim amount) and assign zeros to enrollees with no claims during their spell. Within the 485-day window around the seventieth birthday used for total claimed (Section 3.1), the sample contains 127,359 spells. Just below the threshold, claims are common but modest: 49.8 percent of spells record at least one claim, the average spell generates 2.52 claims, and total claims average NPR 5,310 per spell (mean claim amount, zero for non-claimers, NPR 1,076). Table 2 reports summary statistics for the estimation sample on both sides of the threshold.

Table 2 — Summary Statistics for the Estimation Sample

	Below 70		70 and above	
	Mean	SD	Mean	SD
<i>Panel A. Claims outcomes</i>				
Any claim	0.498	0.500	0.373	0.484
Number of claims	2.52	4.09	1.70	3.41
Total claimed (NPR)	5,310	11,081	3,779	9,674
Mean claim (NPR)	1,076	2,278	841	2,105
<i>Panel B. Enrollee characteristics</i>				
Female	0.519	0.500	0.520	0.500
Age (years)	68.8	0.4	70.3	0.4
Household size	5.23	2.02	2.64	2.46
Non-poor (full premium)	0.865	0.341	0.319	0.466
Ultra-poor (exempt)	0.023	0.151	0.011	0.102
Spells (N)	40,802		86,557	

Note: The sample is all enrolment spells starting May 2023–May 2024 within the 485-day window around the seventieth birthday, the MSE-optimal bandwidth for total claimed in Table 5, $N = 127,359$. Each observation is a spell; claims outcomes are zero for non-claimers. Household size counts members co-enrolled in the spell year and is formed before the age window is applied.

2.4.2. Census data. I supplement the administrative data with the 2021 Nepal Census. The administrative records describe who enrolls, while the census describes the population they enrol from regardless of insurance status. Table 2 is conditional on enrolment, so the two sides of the threshold are expected to differ, because who enrolls responds to the waiver and that response is what the paper measures. What the design requires is that the underlying eligible population not differ across the cutoff, and only a source that observes the uninsured can establish that. The census does two things here. It supplies the covariates for the balance test of Section 3.2, since the register holds little beyond age, gender, and household composition, while literacy, caste, religion, marital status, disability, economic activity, and household wealth are observed only in the census. It also provides an independent test that the eligible population is continuous at 70, reported in Appendix A.3.

Population denominators come from United Nations single-year-of-age estimates for Nepal, which carry the census forward to the sample period. The census is a November 2021 snapshot while the sample covers 2023–2024, so census counts would misdate the population at risk by two years at ages where mortality is steep. The United Nations series is interpolated across single years of age, and every population denominator in the paper comes from it.⁴

3 — Research Design and Validity

3.1. Regression Discontinuity Design

I use the premium waiver at age 70 to estimate what a fall in the price of insurance does to the demand for coverage and to the cost of the pool that takes it up. The design is a local linear regression discontinuity in age at enrolment, with both the intercept and the slope allowed to

4. A second reason to prefer the interpolated series is that the census asks respondents their age rather than their date of birth, so reported ages cluster disproportionately at multiples of five, age 70 among them. The United Nations series carries no such spike. Age heaping therefore matters only for the census microdata, where it affects which individuals fall on either side of the cutoff in the balance test. Appendix A.1 sets out the correction. It leaves the running variable of the main design untouched, since that is computed from date of birth in the administrative register.

differ on each side of the threshold. The unit of observation is the enrolment spell. I estimate

$$Y_i = \alpha + \tau D_i + f(A_i) + D_i \cdot f(A_i) + \varepsilon_i, \quad (1)$$

where Y_i is the spell-level outcome, A_i is age at the spell start date measured in days and centred on the seventieth birthday, $D_i = \mathbf{1}[A_i \geq 0]$ indicates eligibility for the free premium, and $f(\cdot)$ is linear in A_i . The coefficient of interest is τ . Treatment is sharp, because every spell starting at age 70 or above receives the waiver and none below does. Standard errors are clustered at the household level, the level at which premiums are owed and enrolment is decided. Appendix D.2 reports the same estimates with the standard errors clustered instead on the value of the running variable.

Since the data cover only those who enrol, the choice of estimand matters. The parameter τ estimates the difference in claims outcomes between the pool that enrolls at a positive premium just below 70 and the pool that enrolls at a zero premium just above.⁵ It is a *pool-composition contrast*, not the causal effect of free insurance on a fixed population’s utilisation. The enrolled population itself changes discontinuously at the threshold, with population coverage rising from 22 to 42 percent (Section 4), and that compositional response is the object of interest. A price change in a voluntary insurance market operates exactly by changing who buys, so τ measures what that re-sorting does to the risk pool, which is the quantity that determines the programme’s per-enrollee cost and the sustainability of its budget.

Equation (1) deliberately includes no covariates and no fixed effects. Because τ is the total difference between the pool enrolling just below 70 and the pool enrolling just above, conditioning on characteristics that shift at the cutoff would absorb exactly the variation the design is built to measure. The running variable is age in days, measured exactly from the date of birth in the register. It takes 971 distinct values inside the estimation window, so the continuity-based design is valid and no correction for a discrete running variable is needed (Kolesár and Rothe 2018).

I select each outcome’s bandwidth to be mean-squared-error optimal, and I report the robust bias-corrected inference proposed by Calonico, Cattaneo, and Titiunik (2014). For total claimed, the outcome on which the selection analysis of Section 5 rests, the bandwidth is $h = 485$ days.

5. The ultra-poor and other premium-exempt categories of Section 2 face no price change at 70 and remain in the estimation sample on both sides of the threshold.

Within the window I fit local linear regressions with uniform weights, following Imbens and Lemieux (2008), so the estimator is ordinary least squares on the spells inside the bandwidth. Appendix D.1 varies the bandwidth, the kernel and the polynomial order.

Validity rests on the claim that the eligible population evolves smoothly through age 70, so that the only thing changing discontinuously at the threshold is the insurance contract. The main text tests that claim two ways, covariate balance in the census (Section 3.2) and placebo cutoffs at non-policy ages (Section 3.3), and two further checks reinforce it in the appendix: the smoothness of the eligible-population density (Appendix A.3) and permutation checks on inference (Appendix D.2).

3.2. Covariate Balance at the Threshold

The design requires that individuals just above and just below 70 are comparable on characteristics other than the insurance subsidy. Testing this within the enrolled sample would be circular, since enrolment is itself the outcome of the price change, so I test it on the 2021 Nepal Census, which covers the entire population regardless of insurance status.

Census age is self-reported, and respondents disproportionately report ages at multiples of five, so age 70 is both the policy threshold and a heaping spike. I handle this two ways. The first drops reported age 70 outright, so the discontinuity is estimated without ever using the heaped cell (Barreca, Lindo, and Waddell 2016); this assumes nothing about who rounded, but it extrapolates the treated intercept from age 71 upward. The second reassigns the excess at each heaped age to neighbouring ages in proportion to a smooth density fitted on non-heaped ages, repeats that $M = 50$ times, and combines the draws with Rubin’s rules (Heitjan and Rubin 1990; Rubin 1987), so the standard errors carry both sampling uncertainty and the uncertainty from the correction itself. Appendix A.1 details the procedure. The two rest on different assumptions, so agreement between them says more than either says alone.

Table 3 reports both for nine covariates spanning demographics (sex, caste, religion, ever-married status, disability, household size) and economic characteristics (economic inactivity, literacy, wealth quintile). Under the heaping correction no covariate shows a significant discontinuity, and the smallest individual p -value across the nine is 0.26, for literacy. The tests are jointly

null as well: a Wald test that all nine discontinuities are zero, combined across the 50 draws, gives $F(9, 9,941) = 0.65$ with $p = 0.751$. The donut agrees on seven of the nine.

It does not agree on literacy or on the share not working because of age. Both run between two and three percentage points under the donut and are significant at 5 percent. Neither is evidence of a discontinuity in the eligible population. They are the two covariates with the steepest age gradients, and dropping age 70 leaves a support that is still alternately heaped and unheaped, so a local linear fit through three ages on each side can manufacture a jump. Literacy is the clearer case: it is $+0.023$ under the donut and -0.009 under the correction, and a sign reversal of that kind points to residual heaping rather than to anything happening at 70. Running the same donut at ages where no policy changes produces jumps in these two covariates that are larger than the one at age 70 and that alternate in sign, and fitting a quadratic removes the literacy jump entirely. Appendix A.2 plots each covariate against age, and every profile passes through 70 without a visible step.

Table 3 — Covariate Balance at Age 70: 2021 Nepal Census

Covariate	Donut			Heaping-corrected		
	Mean	RD Est.	SE	Mean	RD Est.	SE
Female	0.505	−0.009	(0.011)	0.508	0.007	(0.009)
High Caste	0.476	−0.014	(0.011)	0.477	−0.009	(0.009)
Hindu	0.807	−0.008	(0.009)	0.810	0.004	(0.007)
Ever married	0.988	0.000	(0.002)	0.989	0.000	(0.002)
Has disability	0.064	0.003	(0.005)	0.064	−0.002	(0.004)
Household size	5.153	−0.111	(0.172)	5.203	0.121	(0.134)
Economically inactive (aged)	0.364	0.026**	(0.011)	0.374	−0.001	(0.009)
Literate	0.341	0.023**	(0.011)	0.328	−0.009	(0.008)
Wealth quintile	2.880	0.026	(0.036)	2.855	−0.024	(0.027)
Bandwidth (years)		3.65		3.41		
Observations		80,230		110,292		
Estimator		Local linear (OLS), uniform kernel				

Notes: Each cell is a local linear regression with a uniform kernel, weighted by the census individual weight, with standard errors clustered at the household level. Both bandwidths are selected on the sex indicator; because reported age is integer valued, both select the ages 67 to 73, the donut omitting 70. The columns differ in inference: donut p -values are the robust bias-corrected values of Calonico, Cattaneo, and Titiunik (2014), while the heaping-corrected p -values are conventional and pooled across the 50 draws. Mean Pre-70 is the weighted mean below the cutoff. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.3. *Placebo Cutoffs*

If the claims discontinuity at 70 reflects the insurance-price change rather than some generic feature of utilisation at older ages, the same specification should find nothing at ages where the insurance contract does not change. I therefore re-centre equation (1) at every whole year of age from 64 to 76 and re-estimate, holding the bandwidth, the kernel and the clustering fixed at their values in the headline specification. Ages 69 and 71 are excluded because a symmetric window at either would cross 70 and pick up the real discontinuity, and each remaining placebo is estimated on spells from its own side of the threshold so that no placebo window overlaps the true one. Figure 2 plots the eleven estimates. The exercise runs on total claimed, the outcome the selection analysis of Section 5 rests on.

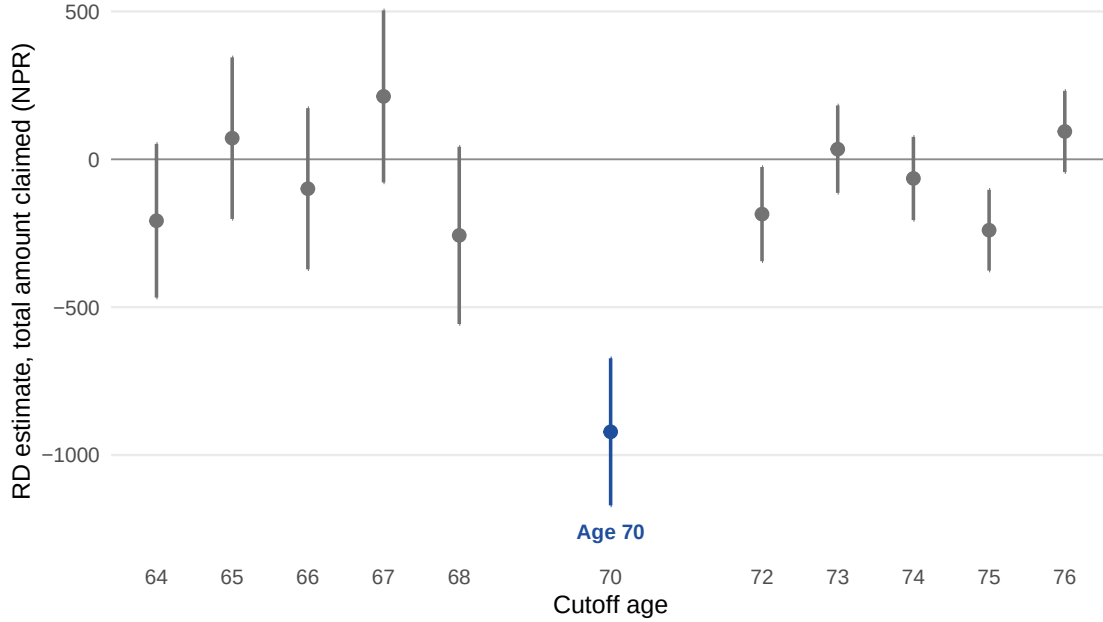
Only at 70 does the design find a discontinuity. Total claimed falls by NPR 921.5 there, a 17.4 percent decline against the just-below mean of NPR 5,310. Across the ten placebo ages the estimates run between $-$ NPR 257 and $+$ NPR 213, between 3 and 5 percent of the same mean, and they alternate in sign rather than trending with age. The largest of them in absolute value is under three-tenths of the estimate at 70, and the age-70 interval sits clear of every other.

Two placebo intervals do exclude zero, at ages 72 and 75. Both sit above the threshold, which is where the design has most power, because coverage jumps at 70 and the placebos above it therefore rest on roughly twice as many spells as those below. Their intervals are correspondingly tighter, so a jump of a few percent of the mean is detectable there while the same jump below 70 would not be. Neither is large. The sweep is better read as a distribution than as ten separate tests, and on that reading the two exceptions establish the scale of ordinary fluctuation rather than undercutting the claim. Nepal's old-age allowance begins at 68 and appears in the sweep for completeness, though it lies outside the estimation window for total claimed and so cannot contribute to what happens at 70.

3.4. *Interpreting the Discontinuity*

The waiver changes the enrolled pool in two ways. Some people would enrol at either price, and for them the waiver removes a premium they were willing to pay. Others enrol only because the price falls to zero, so they appear in the pool above the threshold and not below it. The

Figure 2 – The discontinuity appears only at age 70



Note: RD estimates for total claimed with equation (1) re-centred at each whole year of age, with 95 percent confidence intervals from standard errors clustered by household. Ages 69 and 71 are omitted because a symmetric window at either crosses age 70. The bandwidth is held at $h = 485$ days, the value selected at the true cutoff, so that the placebos differ from the age-70 estimate only in where they are centred. Placebo cutoffs below 70 are estimated on below-70 spells and those above 70 on above-70 spells; the age-70 estimate uses both sides. The above-70 placebos rest on roughly twice as many spells as those below, because coverage jumps at the threshold, so their confidence intervals are visibly tighter.

discontinuity is the sum of what happens to each group, $\tau = \tau_S + \tau_M$, where τ_M is the change in utilisation among those who would have enrolled at either price and τ_S is the change in who is enrolled. Enrolment type is not observed, so the data do not separate the two terms. The institutional design, however, pins down τ_M .

How much care an enrollee uses depends on what that care costs at the point of use, not on the premium, which is already paid by then (Einav and Finkelstein 2018). NHIP charged no copayment or coinsurance over the study period, and the benefit package is uniform on both sides of the threshold (Section 2). Care is therefore free at the margin whether the enrollee is 69 or 70, and ex-post moral hazard has nothing to act on among those who would have enrolled anyway. Two channels remain. The coverage unit switches from a household ceiling to an individual one, which lowers the price of care in the states where the household ceiling would otherwise have bound; ceilings are slack for almost every enrollee on both sides, so this channel is close to zero.

The waived premium is also a transfer of NPR 3,500 to the household, and if care is a normal good that transfer weakly raises utilisation. Both point the same way and both are small, so τ_M is approximately zero and weakly positive.⁶

The compositional response turns on a single object: the expected cost of the enrollees a zero price draws in, measured against the pool already paying (Einav and Finkelstein 2011). If the marginal enrollees are cheaper, average claims fall. This is the classical adverse-selection case (Akerlof 1970; Rothschild and Stiglitz 1976). If they are more expensive, average claims rise; this is advantageous selection, in which those who paid the premium were the risk-averse and the healthy (Fang, Keane, and Silverman 2008). Expected cost itself has two parts. It reflects how sick a person is, and it reflects how much more care that person would use once covered. Enrollees can sort on either. Einav et al. (2013) show that sorting on the second, selection on moral hazard, is a component of adverse selection rather than a force beside it, because both parts enter expected cost in the same way. Nothing in the theory fixes the sign.

The identity constrains what τ_S can be. If τ is negative then $\tau_S = \tau - \tau_M$ is at least as large in magnitude as τ , because $\tau_M \geq 0$ works against a fall, so the composition term must more than account for it. A fall in claims is therefore a lower bound, in magnitude, on the compositional channel, and not a net of two forces that could understate it. Where advantageous selection also operates it pushes that term the other way, so the classical channel must be larger still.

This is what makes the sign worth reporting on its own. Claims falling where free coverage might have been expected to raise them is consistent with the marginal enrollees being substantially cheaper than the pool that was already insured. It does not establish how much cheaper, and it does not price the pool. Section 5 prices the pool from realised costs, and Section 6 asks what else could produce the decline.

6. One channel runs the other way. Enrollees below 70 paid the premium while those above did not, so a sunk-cost response would raise utilisation on the paying side and make τ_M negative. Single-person households are the cleanest check, because their coverage terms are identical across the threshold and only the price changes.

4 — Results

4.1. Demand for Health Insurance

The waiver's first effect is on demand for coverage. Figure 3 plots NHIP coverage, enrollees as a share of the age-eligible population, by single year of age, counting as insured everyone whose enrolment spell covers 25 November 2023. Coverage is flat through the sixties and jumps at 70, from 22 percent just below the threshold to 42 percent just above, a gain of $\Delta Q = 0.20$.⁷ At the margin a fifth of the eligible population takes up coverage at 70 that it declined at NPR 3,500, so the design compares take-up at two administered prices rather than tracing a demand schedule between them.⁸ That jump is what makes the results below statements about pool composition. The group enrolled just above 70 is not the group enrolled just below plus a treatment. It is a larger and differently selected pool, so every contrast across the threshold measures what the re-sorting does to who the programme insures.

Table 4 asks who the zero price draws in, reporting the age-70 discontinuity in enrollee characteristics rather than in outcomes.⁹ The waiver reaches past the pool already paying. Enrollees at the threshold live 0.54 km farther from their assigned first point of contact, 7.1 percent above the 7.57 km faced by those enrolling just below, and they have 0.74 fewer empanelled facilities within 10 km of their ward, 12.7 percent below a baseline of 5.81. They are 3.0 percentage points less likely to live in a municipality rather than a rural municipality, and the poverty rate of their local unit is 1.7 percentage points higher, 10.1 percent above the 0.163 that prevails below the cutoff.

The pool also shifts on characteristics carried by the person rather than the place. The share of enrollees from the Khas-Arya Brahman and Chhetri group falls by 1.8 percentage points, 4.1 percent against a base of 0.428, while the share who are women rises by 1.6 percentage points on

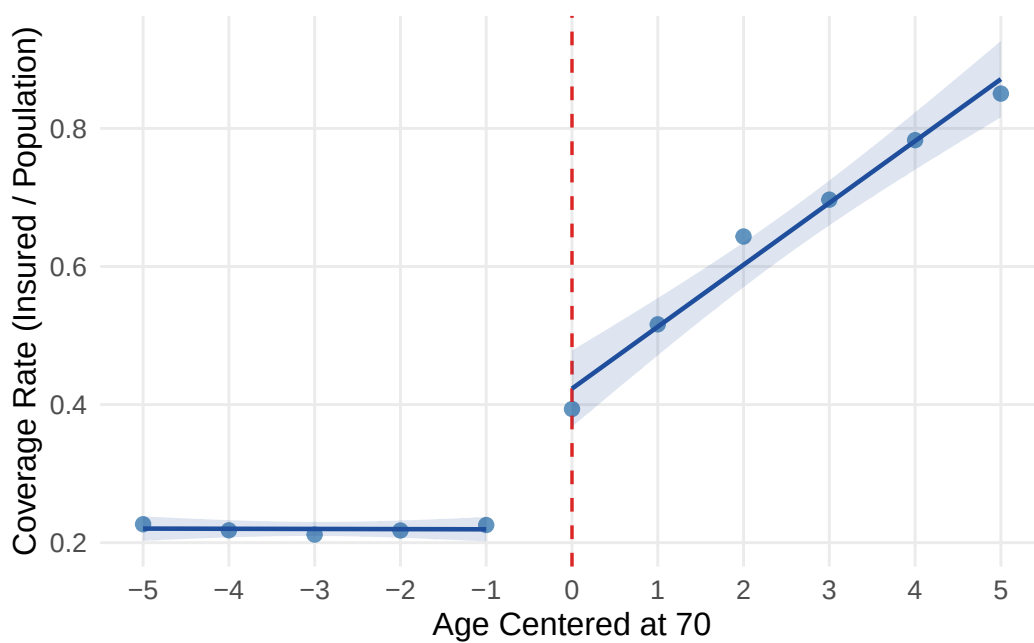
7. The gain is a lower bound. Enrolment windows are quarterly and the premium is waived only from the first start date at which an enrollee is 70, so the age-70 group still contains people who have turned 70 but have not yet reached a window at which the waiver applies.

8. The price that falls is the statutory premium on a marginal individual policy. Below 70 an individual whose household is not already enrolled can obtain coverage only by enrolling the household at NPR 3,500, while at 70 that individual can enrol alone at no charge.

9. Distance is to the assigned first point of contact, the nearest public facility by rule. The facility count is measured at the ward centroid, so it is common to everyone in a ward. Urban denotes any local unit other than a rural municipality. Local poverty is the headcount for the enrollee's local unit.

a base of 0.519.¹⁰ Both movements are smaller than the geographic ones and neither is significant at the 1 percent level, so they are best read as consistent with the same pattern rather than as independent evidence for it. The programme is meant to extend coverage to elderly Nepalis who would not otherwise buy it, and on its own terms the waiver reaches that margin. The enrollees it adds come from poorer, more rural and less well served places than those already paying at 69, which is where the coverage gap was. Whether they also use the coverage is a separate question, and Section 4.2 takes it up.

Figure 3 – Coverage jumps at age 70



Note: NHIP enrollees as a share of the United Nations single-year population estimate for Nepal, measured at 25 November 2023. A person counts as insured if an enrolment spell covers that date. Recorded counts are adjusted for deaths not retired from the register; the adjustment leaves ages 70 and below unchanged and so does not affect the discontinuity. The dashed line marks age 70; fitted lines are local linear on each side of the threshold.

10. Caste should be read with a little caution because it is recorded for 80 percent of the sample and recording is itself discontinuous at the threshold. The estimate is still about composition rather than differential recording, because the characteristics observed for every enrollee are unchanged when I re-estimate them on the recorded subsample.

Table 4 — RD estimates at the age-70 threshold, by enrollee characteristic

	Individual				Local area	
	Distance to first contact (km) (1)	Facilities within 10 km (count) (2)	High caste (0/1) (3)	Female (0/1) (4)	Urban municipality (0/1) (5)	Local poverty rate (share) (6)
RD estimate	0.536*** (0.116)	−0.740*** (0.101)	−0.0177* (0.0070)	0.0155** (0.0060)	−0.0303*** (0.0052)	0.0166*** (0.0014)
R. <i>p</i> -value	<0.001	<0.001	0.055	0.016	<0.001	<0.001
Mean	7.57	5.81	0.428	0.519	0.756	0.163
% change	7.1%	−12.7%	−4.1%	3.0%	−4.0%	10.1%
BW (days)	485	485	485	485	485	485
N	125,859	127,359	101,502	127,359	126,601	127,359

Notes: Each observation is an enrolment spell. The running variable is the enrollee’s age at the start date of the spell, measured in days and centred on the seventieth birthday. Estimates are local linear with separate slopes on each side of the cutoff and a uniform kernel, fitted by ordinary least squares with no covariates and no fixed effects. Every column is estimated at a common bandwidth of 485 days, the mean-squared-error-optimal choice of Calonico, Cattaneo, and Titiunik (2014) for claim value, so all columns rest on one window. Standard errors in parentheses are clustered at the household level; the reported *p*-value is the robust bias-corrected *p*-value. The estimation sample is 127,359 spells; columns 1, 3 and 5 are defined for 99, 80 and 99 percent of it respectively and every other column covers it in full. Local poverty is from the 2023 small-area estimation (National Statistics Office 2026). Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.2. The Cost of Coverage

Table 5 reports claims over the twelve months of the enrolment spell, for the pool that enrolls free against the pool that pays. I focus on four indicators: the total value of claims; the mean claim; whether an individual had a claim, overall, chronic or inpatient; and the number of claims, overall, chronic or inpatient. I show the discontinuity in Figure 4, and Appendix B.1 plots the remaining two margins, which step down at the threshold in the same way.

Across all the outcomes cost per enrollee falls at the threshold. Total claims per spell fall by NPR 921.5 against NPR 5,310, a decline of 17.4 percent, and the mean claim by NPR 161.6 against NPR 1,081, a decline of 15.0 percent (columns 1 and 2). The probability of any claim falls by 5.9 percentage points against a below-cutoff mean of 0.497, a decline of 11.8 percent (column 3). The number of claims falls by 0.22 against 2.56, a decline of 8.7 percent (column 6). All four are

Table 5 — RD estimates at the age-70 threshold, by claims margin

	Claim value		Had a claim			Number of claims		
	Value (NPR) (1)	Mean claim (NPR) (2)	Any (0/1) (3)	Chronic (0/1) (4)	Inpatient (0/1) (5)	Any (count) (6)	Chronic (count) (7)	Inpatient (count) (8)
RD estimate	−921.5*** (126.9)	−161.6*** (21.9)	−0.059*** (0.007)	−0.048*** (0.007)	−0.0128*** (0.0024)	−0.223*** (0.057)	−0.144*** (0.038)	−0.0204*** (0.0039)
R. <i>p</i> -value	<0.001	<0.001	<0.001	<0.001	<0.001	0.001	0.001	<0.001
Mean	5,310	1,081	0.50	0.36	0.075	2.56	1.39	0.105
% change	−17.4%	−15.0%	−11.8%	−13.2%	−17.0%	−8.7%	−10.4%	−19.4%
BW (days)	485	764	342	342	794	312	312	840
N	127,359	205,305	88,148	88,148	214,701	81,242	81,242	228,041

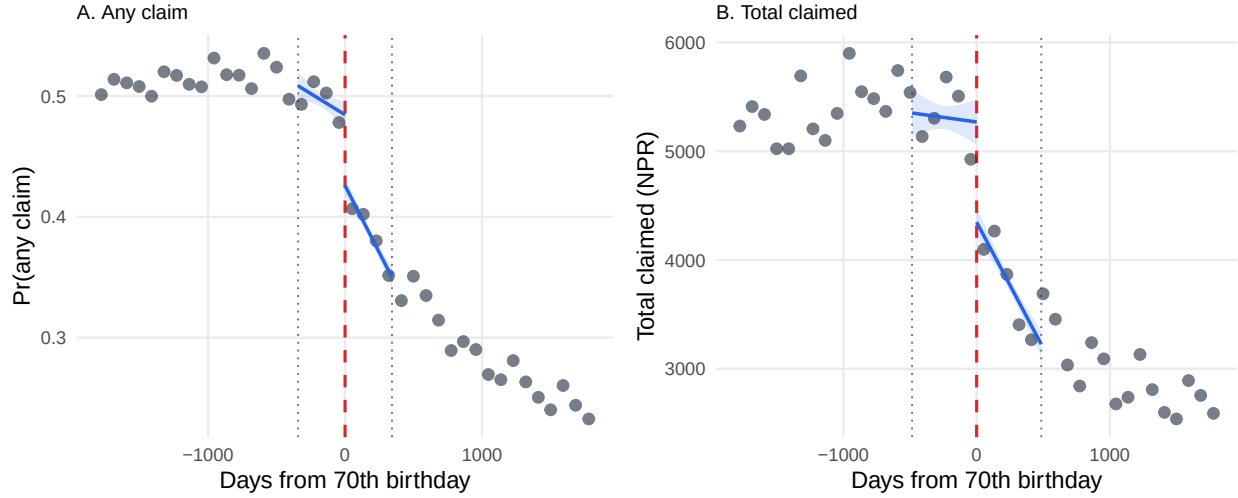
Notes: Each observation is an enrolment spell. The running variable is the enrollee’s age at the start date of the spell, measured in days and centred on the seventieth birthday, so it is zero for a spell beginning on that birthday. Estimates are local linear with separate slopes on each side of the cutoff and a uniform kernel, fitted by ordinary least squares with no covariates and no fixed effects. Each bandwidth is the mean-squared-error-optimal choice of Calonico, Cattaneo, and Titiunik (2014) for the outcome, with one exception: the chronic column of each block is estimated at the bandwidth selected for its overall column, so columns 3 and 4, and columns 6 and 7, each rest on one sample and can be compared. Standard errors in parentheses are clustered at the household level; the reported *p*-value is the robust bias-corrected *p*-value. Every column is defined for the whole estimation sample, non-claimers included. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

significant at the 1 percent level. Overall, fewer enrollees claim, those who claim do so less often, and their claims are smaller. Free health insurance brought in lower-cost enrollees.

The decline holds wherever the new enrollees live. Appendix B.4 re-estimates it within bands of distance from the enrollee’s ward to the registered first point of contact, and no band differs significantly from any other.

The pool that pays also claims sooner, which Table B2 in the appendix sets out. The probability of a claim within 30 days falls by 4.9 percentage points against a below-cutoff mean of 0.225, a decline of 21.9 percent, and the probability of a claim within 90 days by 5.7 percentage points against 0.369, a decline of 15.5 percent. Both fall by proportionally more than the probability of a claim at any point in the spell, so what goes at the threshold is disproportionately the enrollee who claims early. Days from the start of the spell to the first claim, among enrollees who claim, rise by 7.4 against a mean of 67.7, an increase of 10.9 percent; that column conditions on claiming,

Figure 4 – The discontinuity in claims at age 70



Note: Binned means in quarterly bins of age at the spell start date, shown over twenty quarters on each side of the seventieth birthday, as in Figure 4 of the paper. The dashed vertical line is the cutoff; the dotted lines mark the estimation bandwidth, which is selected separately for each outcome and is 342 days in Panel A and 485 in Panel B. Local linear fits are estimated on each side of the cutoff on the spell-level data *inside the bandwidth only*, with a uniform kernel and no covariates or fixed effects, so the gap drawn at the cutoff is the coefficient reported in Table 5. Nothing is extrapolated: no fit is drawn beyond the bandwidth, and the bins outside it are shown for context and are descriptive only. Bins are a display device and no reported quantity depends on the bin width. Shaded bands are 95 percent confidence intervals from standard errors clustered on the household.

so the claimant pool itself changes at the cutoff and it is descriptive. Time to first claim proxies for the value of a claim with greater precision than the value itself (Aron-Dine et al. 2015), and someone who buys coverage knowing she needs care claims quickly.

4.3. The Health of the Enrollees

The enrollees who join for free are not only less costly, they are also healthier on average. I show that on two margins: treatment for a chronic condition (columns 4 and 7) and admission to hospital (columns 5 and 8). I plot the discontinuity for both in Figure 5.

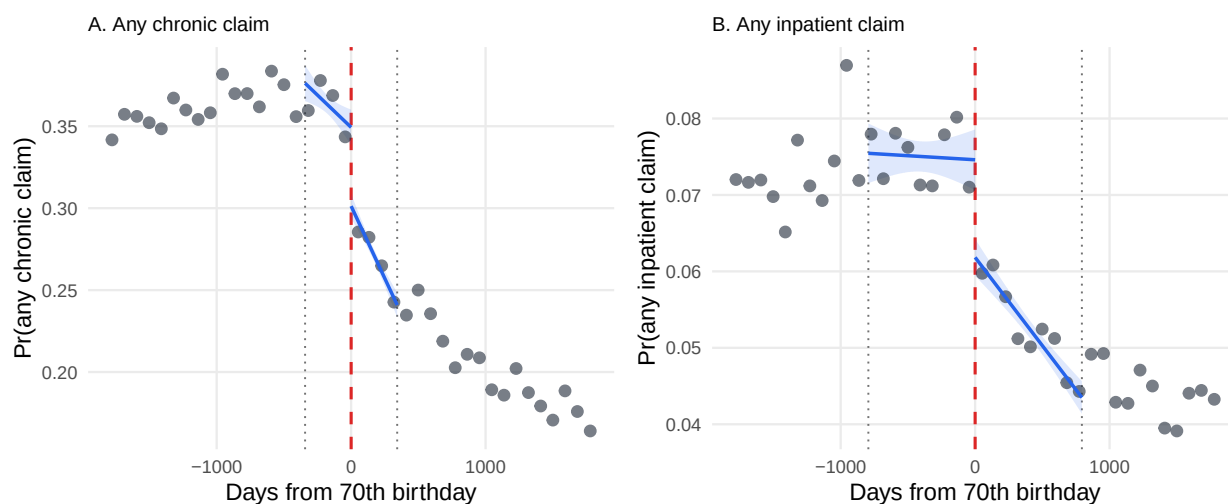
Chronic conditions are the closest thing to a pre-determined health state a claims dataset contains, because nobody acquires diabetes in response to a premium waiver. I classify a claim as chronic from its ICD-10 code, following Banerjee et al. (2021).¹¹ Chronic claims are 53.7 percent

11. They build the classification from the Chronic Condition Indicator of the Healthcare Cost and Utilization Project, which scores every diagnosis code of ICD-10-CM, the American clinical modification, as chronic or not (Agency for Healthcare Research and Quality 2018). The indicator is then collapsed to the three-digit stem of the code, and a diagnosis counts as chronic when more than 75 percent of the codes in its stem are chronic. The stem is

of claims in this population, with hypertension alone at 15.0 percent and type-2 diabetes at 10.0 percent, which is what an elderly outpatient caseload looks like. They fall by proportionally more than claims of any kind. The probability of a chronic claim falls 13.2 percent against 11.8 percent (column 4), and the number of chronic claims falls 10.4 percent against 8.7 percent (column 7). Chronic claims are 54.6 percent of the below-cutoff claim count but 64.7 percent of the fall in it, so what thins at the threshold is disproportionately the management of ongoing disease.

I find the same effect on admission to hospital. An inpatient stay clears a clinical threshold that an outpatient visit does not, and it is not something an enrollee can award herself. The probability of an inpatient claim falls by 1.3 percentage points against a below-cutoff mean of 0.075, a decline of 17.0 percent (column 5), and the number of inpatient claims falls by 0.020 against 0.105, a decline of 19.4 percent (column 8). Both are significant well below the 1 percent level. These are the largest proportional declines in the table, ahead of the 17.4 percent fall in total claims and well ahead of the 11.8 percent fall in the probability of claiming at all.

Figure 5 – The discontinuity on the two health margins



Note: Constructed as Figure 4. Panel A is drawn at the bandwidth selected for any claim, 342 days, rather than at its own, which is the pairing rule of Table 5; Panel B is drawn at its own bandwidth of 794 days, so the two panels display the same twenty quarters of data but are estimated on different windows. The gap drawn at the cutoff is the coefficient reported in columns 4 and 5 of that table.

the unit because it is the finest level at which ICD-10-CM and the World Health Organization's ICD-10 correspond, and the scored classification is the American one while the recorded diagnoses are not: Indonesia's scheme and NHIP both code in the WHO classification, so their mapping problem is also mine and the rule transfers without adaptation. Of the 1,910 stems in the indicator, 583 clear the threshold. I use the 2018 vintage, which is the one they cite and which scores codes as chronic or not; the indicator has since been revised to add a third category for codes with no determination, which would change the denominator of the 75 percent rule.

The pool that pays is therefore in worse health, on two measures moral hazard does not move. Neither is a baseline measure of health: each records a diagnosis that was presented, so each is post-treatment, and I report them as margins of the outcome rather than as controls or in a balance test. The evidence is consistent with adverse selection; it does not by itself establish it.

5 — Adverse Selection at the Age-70 Threshold

Section 4.2 showed that claims per enrollee fall at the threshold, and Section 3.4 showed that the fall is a change in who is enrolled rather than in how anyone behaves. What it leaves open is what the new enrollees cost. They are never observed on their own. They appear only inside the average above the threshold, mixed with the people who would have enrolled at either price, and it is these two groups that a claim about selection has to compare. This section separates them. I use the waiver as a source of variation in the composition of the insured pool, estimate what the enrollees it draws in cost, and compare that with what the enrollees who pay cost.

I adapt the framework of Einav and Finkelstein (2011) to make that comparison. NHIP defines one contract by statute, and the choice facing an eligible person is binary, to enrol or not. Selection therefore operates on participation, the margin Akerlof (1970) describes, rather than on sorting across a menu of coverage levels (Rothschild and Stiglitz 1976).¹² Finkelstein, Hendren, and Shepard (2019) use the same strategy at premium discontinuities in the Massachusetts subsidized exchange, recovering the cost of the enrollees each price change moves. I apply it at one discontinuity in place of their three. What changes at 70 is the price and the unit at which the benefit ceiling applies, not the number of contracts on offer; the change in the ceiling is taken up as a condition on the calculation in Section 5.4.

I find evidence of adverse selection into enrollment under the paid regime. The enrollees the waiver draws in, roughly a fifth of the age-eligible population, generate NPR 3,358 each in paid claims, compared with NPR 5,269 for the enrollees the paid regime retains: a ratio of 0.637. The marginal enrollees are thus about 36 percent cheaper to cover than the pool they join. This is

12. The single-contract setting also bounds what the exercise can deliver. Selection can distort which contracts are offered at all, a margin Einav and Finkelstein (2011) note may carry a substantial part of the welfare consequences of selection. In NHIP the contract is fixed by statute rather than set by competing insurers, so the omitted margin is genuinely absent rather than assumed away.

the configuration Finkelstein, Hendren, and Shepard (2019) interpret as adverse selection at their premium discontinuities.

5.1. Setup and Assumptions

Consider a population of eligible individuals who may buy a single insurance contract at price p . Index them by i , with willingness to pay w_i for the contract and expected cost c_i of insuring them. Demand at price p is the share who buy, $D(p) = \Pr(w_i \geq p)$, and I write $Q = D(p)$ for coverage. Ordering the eligible population by willingness to pay allows both cost curves to be written as functions of coverage: $MC(Q)$ is the expected cost of the individual at coverage rate Q , the last person a price of p brings in, and $AC(Q)$ is the average cost of everyone insured at that coverage rate. The two are linked by construction, because total cost is the running sum of what the marginal enrollees cost:

$$AC(Q) Q = \int_0^Q MC(q) dq, \quad \text{so} \quad AC'(Q) = \frac{MC(Q) - AC(Q)}{Q}. \quad (2)$$

Average cost falls exactly where marginal cost lies below it.

The argument rests on four assumptions.

ASSUMPTION 1 (Monotonicity): *No one who enrolls at the statutory premium would decline coverage at a price of zero.*

ASSUMPTION 2 (Continuity): *The eligible population is continuous at the threshold, so the pools on either side are drawn from the same underlying group.*

ASSUMPTION 3 (Cost): *The cost of insuring a person is the claims the programme pays on their behalf.*

ASSUMPTION 4 (No Differential Behavioural Response): *Crossing the threshold does not change how a given person uses care, so a change in measured cost reflects a change in who is insured.*

Assumption 3 follows Finkelstein, Hendren, and Shepard (2019), who define insurer cost as medical claims paid and abstract from administrative costs.¹³ All four, together with the contract-

13. Fischer, Frölich, and Landmann (2023) instead build an expected-cost index from the detailed baseline covariates their experiment collected before enrolment. They have two reasons: their claims data record only 39 claims, so claim-based cost curves would be too imprecise, and an index built from baseline values cannot absorb any utilisation response to coverage. Their index is estimated from 334 inpatient events reported in a high-frequency phone survey.

equivalence condition of Section 5.4, are what the calculation of the next subsection requires. Assumption 4 in particular is not merely interpretive, because without it the difference in total costs across the threshold mixes the compliers' claims with a change in what the already-insured cost.

5.2. What the Waiver Identifies

The waiver delivers the insured share and the average cost of the insured pool on each side of the threshold. Below it, where coverage requires the statutory household premium, a share Q_{below} of the age-eligible population is insured and that pool costs AC_{below} per enrollee; above it, where coverage is free and individual, the share is Q_{above} and the pool costs AC_{above} . The cost of the marginal enrollees can then be derived from average costs and demand by straightforward calculation. Applying equation (2) at the two coverage rates and differencing gives the total cost of everyone in between,

$$AC_{\text{above}} Q_{\text{above}} - AC_{\text{below}} Q_{\text{below}} = \int_{Q_{\text{below}}}^{Q_{\text{above}}} MC(q) dq, \quad (3)$$

and dividing by the mass the waiver draws in, $\Delta Q = Q_{\text{above}} - Q_{\text{below}}$, gives their average cost:

$$\overline{MC} \equiv \frac{1}{\Delta Q} \int_{Q_{\text{below}}}^{Q_{\text{above}}} MC(q) dq = \frac{AC_{\text{above}} Q_{\text{above}} - AC_{\text{below}} Q_{\text{below}}}{\Delta Q}. \quad (4)$$

Every term on the right-hand side is estimated from the design, so under the four assumptions, together with the contract-equivalence condition examined in Section 5.4, $\overline{MC}$ is the average cost of the enrollees whose insured status changes at the threshold. $MC(\cdot)$ is a curve and $\overline{MC}$ is a number, the average of that curve over the interval the waiver induces. The design delivers the second, not the first, so the slope of $MC(\cdot)$ is not identified.

The two averages are nonetheless enough to compare the marginal group with the infra-

Neither reason transfers cleanly. Precision is not my binding constraint, since the claims data cover 32.9 million reimbursements; but the registration records carry no comparable baseline health information, and the enrollees the waiver draws in enter the data only once insured, so no pre-enrolment history exists from which an index of their kind could be built. Realised claims are therefore the feasible cost measure here as well as the convention of Finkelstein, Hendren, and Shepard (2019), at the price that the contamination their index design rules out must instead be signed by argument, which is what Assumption 4 and Section 5.4 undertake.

marginal one. Since the insured pool above the threshold is the pool below plus the enrollees the waiver adds, AC_{above} is a weighted average of AC_{below} and $\overline{MC}$ and must lie between them. Average cost therefore falls at the threshold if and only if

$$\overline{MC} < AC_{\text{below}}, \quad (5)$$

so a fall in average cost is equivalent to the statement that the enrollees the waiver adds are cheaper than the enrollees the paid regime retains. This is the contrast Finkelstein, Hendren, and Shepard (2019) report at each of their premium discontinuities, and it is what I mean by adverse selection in what follows.

5.3. Results

The enrollees the waiver draws in cost NPR 3,358 each, against NPR 5,269 for the pool that was already paying: a ratio of 0.637, and a gap of NPR 1,911 per enrollee. The one comparable estimate, at a premium discontinuity in the Massachusetts subsidized exchange, is 0.59 (Finkelstein, Hendren, and Shepard 2019).¹⁴ $\overline{MC}$ lies below AC_{below} in all 500 bootstrap draws. The inputs are in Table 6. Coverage rises from 0.219 of the eligible population just below the threshold to 0.423 just above it, so the waiver draws in $\Delta Q = 0.204$, roughly a fifth of the eligible population, and average cost among the insured falls over the same interval from NPR 5,269 to NPR 4,348. That decline of NPR 922 comes from the same local-linear limits of realised claims that produce the claims discontinuity in Table 5, so the two exhibits restate one estimate rather than provide two.

Figure 6 plots the two average-cost points, the induced interval between them, and $\overline{MC}$ at its midpoint. The dashed line is the marginal cost implied by assuming average cost is linear in coverage. It is drawn for orientation only: it agrees with $\overline{MC}$ at the midpoint as a matter of

14. Their design does not report this ratio; I construct it from their Figures 7B and 8B, fiscal year 2011. At the 150-percent-of-poverty threshold, enrollment in the composite high-coverage plan falls from 80 to 64 percent of the eligible population as the enrollee premium rises, and average cost among those who remain rises from \$361 to \$393 per month. Equation (4) applied to those four numbers gives \$233 for the enrollees who leave, against the \$393 average of the pool that stays. Two things keep this from being a match. Their margin is exit from the high-coverage plan into either the low-coverage plan or uninsurance, a composite they note makes marginal-cost language less useful, whereas mine is insurance against none; and the cost jump that carries the ratio has a standard error of \$16.7 against a point estimate of \$32.3. Finkelstein, Hendren, and Shepard (2019) themselves headline marginal cost of \$333 against average cost of \$417, or 0.80, at the *median* of their willingness-to-pay distribution; that object conditions on a point of the type distribution rather than on a threshold contrast, and is not the analogue of my estimate.

Table 6 — Coverage and costs at the age-70 threshold

	Point	Boot. SE	95% CI
<i>Panel A: Coverage</i>			
Q_{below}	0.219	0.001	[0.217, 0.222]
Q_{above}	0.423	0.001	[0.420, 0.426]
ΔQ (induced margin)	0.204	0.002	[0.200, 0.208]
<i>Panel B: Costs (NPR per enrollee)</i>			
AC_{below}	5,269	106	[5,077, 5,477]
AC_{above}	4,348	65	[4,237, 4,491]
$AC_{\text{above}} - AC_{\text{below}}$	-922	122	[-1,132, -658]
$\overline{MC}$ (induced enrollees)	3,358	175	[3,050, 3,739]
$AC_{\text{below}} - \overline{MC}$	1,911	255	[1,363, 2,366]
$\overline{MC}/AC_{\text{below}}$	0.637	0.043	[0.563, 0.729]

Notes: Coverage is the point-in-time insured stock of Section 4.1. Costs are realised claims per enrolment spell, taken as local-linear limits on age in days at $h = 485$ days, so $AC_{\text{above}} - AC_{\text{below}}$ is the discontinuity in total claimed reported in Table 5. $\overline{MC}$ is the average cost of the enrollees the waiver draws in, recovered from the identity in the text. Households are resampled with replacement, $B = 500$ times. Each replication uses the same resampled households for the claims and for the insured counts, so the average costs and the coverage shares move together rather than independently. Population denominators and the mortality correction are administrative inputs and are held fixed across draws. Intervals are bias-corrected percentile intervals, and the interval for a ratio is computed from its own bootstrap distribution. $\overline{MC}$ falls below AC_{below} in 100.0 percent of draws. Sensitivity to the bandwidth used for the average costs is in Table C1.

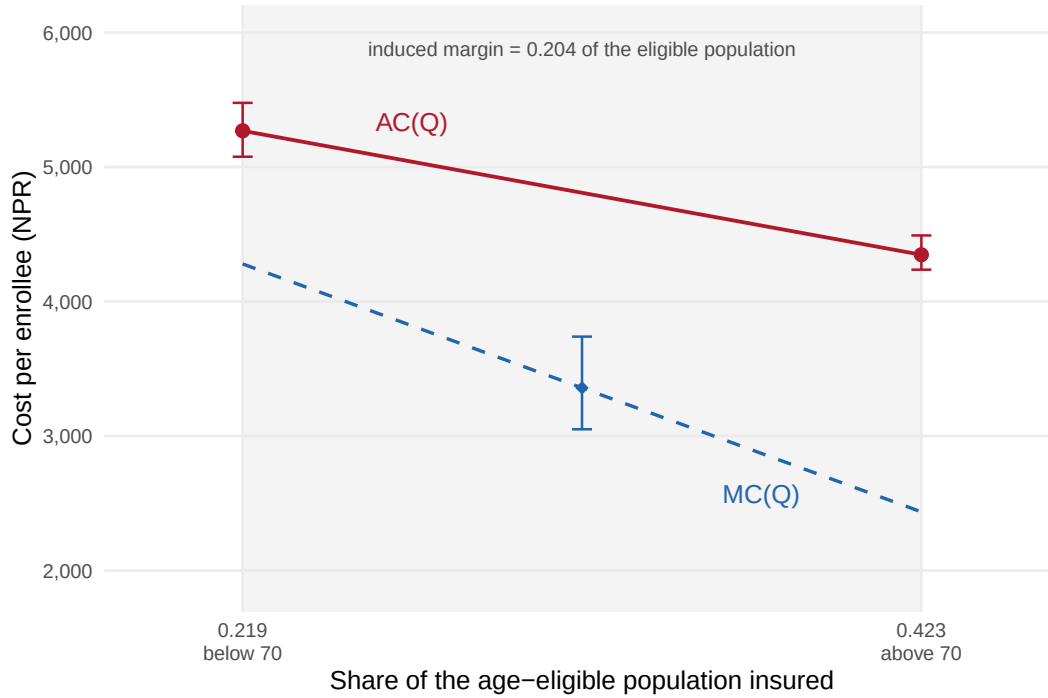
algebra rather than of evidence.

The ratio is sensitive to bandwidth. Across bandwidths of 365 to 1,095 days (Table C1), $\overline{MC}$ moves from NPR 3,726 to NPR 2,924 and the ratio from 0.728 to 0.554. The ordering $\overline{MC} < AC_{\text{below}}$ holds at every bandwidth, but the magnitude of the gap carries the bandwidth sensitivity of any regression discontinuity quantity, and I present the ratio as an estimate with an interval rather than as a constant of the programme.

The four inputs to equation (4) come from two designs. The average costs are local-linear limits of realised claims per enrolment spell; the coverage shares are the point-in-time stocks of Section 4.1. The two differ in population, running variable, bandwidth and treatment of decedents, so $AC \cdot Q$ is not the total cost of one enumerated pool. A denominator error common to both sides cancels: $\overline{MC}$ and the ratio are unchanged when Q_{below} and Q_{above} are scaled by a common factor, so only error that differs at 70 can matter.

These estimates are local. The compliers are individuals near age 70, and the cost-participation relationship I document is theirs; extrapolating the selection pattern to other ages, or to the

Figure 6 — Average and marginal cost over the coverage induced at age 70



Note: Cost per enrollee against the share of the age-eligible population insured. The two average-cost points are the local-linear limits of realised claims per enrolment spell on each side of the seventieth birthday, at $h = 485$ days, plotted at the coverage rates of Section 4.1; they are joined by a segment, as in Finkelstein, Hendren, and Shepard (2019). Bars are 95 percent bias-corrected bootstrap intervals. The diamond is $\overline{MC}$ from equation (4), the average cost of the enrollees the waiver draws in, plotted at the midpoint of the induced interval. The marginal cost line is dashed because its slope is not estimated: two average-cost points pin down a line $AC'(Q) = a + bQ$, and total cost $AC(Q) \cdot Q$ then implies $MC(Q) = a + 2bQ$, a line of twice the slope (Fischer, Frölich, and Landmann 2023). The estimated $\overline{MC}$ and the implied line coincide at the midpoint by construction. Shading marks the induced interval; nothing is drawn outside it.

never-takers who decline free coverage, requires assumptions the design does not test.

5.4. Threats to Interpretation

Assumption 2 is the standard regression discontinuity requirement, and it is defended with the rest of the design in Section 6.3. Assumption 3 is a measurement convention. The burden the model carries falls on the other two, on the contract, and on the unit at which the decision to enrol is taken.

Continuity of the insured pool (Assumption 1). Free coverage weakly dominates paid coverage on both the premium and the benefit ceiling, so no one prefers to be uninsured at 70 having paid at

69. But the assumption the calculation needs is about process, not preference: everyone insured just below the threshold must still be insured just above, which requires that the conversion from a listed household member to an individual policyholder does not fail. If it fails for the frail, expensive payers leave the pool at 70 and their departure mimics the pattern I report. Two pieces of evidence bear on it, and both cut against a systematic failure. Coverage nearly doubles at the threshold, from 0.219 to 0.423 of the eligible population, which is hard to reconcile with a conversion process that drops people on the way through. And among enrollees whose coverage lapses and later resumes, those whose gap spans their seventieth birthday recorded NPR 8,757 in claims before 70 against NPR 19,354 for everyone else (Section 6.3): the enrollees who do leave the pool around the threshold are less than half as expensive as those who do not, the opposite of the pattern that would generate my result. Neither is a direct measurement of one-way exit at 70, so together they bound the threat rather than close it.

The contract change at 70 (contract equivalence). At the threshold the benefit ceiling switches from a household total of NPR 100,000 to an individual entitlement, so the contract is not literally identical on the two sides, and claims above the threshold are recorded under the more generous version. If ceilings bound, AC_{above} would be inflated by the contract change rather than deflated by composition. They do not bind. Because the ceiling below the threshold is shared, the test must be run on household totals: 0.48 percent of household-spells reach 80 percent of the ceiling and 0.09 percent reach it (Table C2). The constraint is close to irrelevant on both sides, so the fall in average cost is not the ceiling relaxing. Consistent with weak enforcement rather than a binding cap, the largest single spell below the threshold records NPR 158,117 in claims, above the nominal ceiling.

Behavioural response (Assumption 4). Because $\overline{MC}$ is built from realised claims, it could in principle reflect how the enrollees the waiver draws in use care rather than how sick they are. The classical channel is absent: care is cashless at the point of service on both sides of the threshold, and copayments were introduced only in 2025, after my sample ends, so crossing the threshold changes the price of coverage but not the price of care.¹⁵ Two channels remain above the threshold. The waived premium relaxes the household budget, and the individual entitlement removes the

15. Statutory basis and dates in Section 2.

shadow price a member of a household faces when her claims draw down an allowance shared with relatives. Both raise the use of care, so both inflate AC_{above} and, through equation (4), inflate $\overline{MC}$: my estimate is an upper bound on the cost of the marginal enrollees net of behaviour, and the finding that it lies below AC_{below} is conservative with respect to them. Only a response in the other direction could threaten the finding, and it would have to be large. Write τ_M for the change, on crossing the threshold, in what an enrollee who would have enrolled at either price costs, and c_E for what the entrants cost. Equation (4) then returns $\overline{MC} = c_E + (Q_{\text{below}}/\Delta Q) \tau_M$, and with $Q_{\text{below}}/\Delta Q = 1.07$, pushing c_E above AC_{below} would require τ_M of about $-1,780$: enrollees just below the threshold spending about a third of AC_{below} more than they would have above it, purely because they had paid a premium.¹⁶

The household as the decision unit (single ordering). Below the threshold the premium is a household bundle: NPR 3,500 covers up to five members, so covering one more member of a household of four costs nothing while an unattached individual faces the full NPR 3,500. No common individual price exists, so the insured pool below the threshold is not the top segment of any individual willingness-to-pay ordering. The comparison can therefore come out as it does with no relation at all between an individual's willingness to pay and her own cost: it suffices that households enrol *because* they contain an expensive elder, which loads the paid pool with high-cost members through the household channel alone. Household composition does change at the threshold, since the mean size of the enrolling household falls sharply once coverage is free and individual. What the estimates deliver is a comparison between the marginal and inframarginal groups the waiver defines, not between two points on an individual cost curve.

6 – Robustness and Threats

This section reports a wide range of robustness and falsification checks on the estimate. I begin with falsification, showing that the discontinuity is policy-induced: it is absent from every cohort

16. This is not a stronger position than Finkelstein, Hendren, and Shepard (2019). Their selection contrast holds the plan fixed on both sides of each premium discontinuity, so it needs no argument about behaviour at all; their appeal to the magnitude of moral hazard defends a different claim, about willingness to pay against cost. It is my contrast that spans a change in the terms of coverage, which is why Assumption 4 and the ceiling evidence have to carry weight here.

that enrolled before the free-premium provision existed. I then show that the results remain robust to the choices a local-linear design requires, across a bandwidth grid, donut variants, kernel weights and polynomial order, and that their precision does not rest on the clustering assumption. The remaining subsections take the threats in turn, covering manipulation of the running variable, strategic timing of re-entry, two rival readings of the compositional shift, and the right-censoring of reimbursement records.

6.1. *Falsification: Pre-Policy Placebo*

The discontinuity is absent from every cohort that enrolled before the free-premium provision existed. NHIP waived the premium at 70 from 14 April 2019, so splitting the sample by enrolment quarter and re-estimating the RD in each one gives a placebo in the eight quarters before that date and an event study across the nineteen after it. Figure 7 plots all four claims margins.

Before the waiver, claims do not move at 70. The total-claimed discontinuity straddles zero across the eight pre-waiver quarters and is never distinguishable from it, and the other three margins are flat over the same window. Three of the thirty-two pre-waiver estimates cross the 5 percent level, which is what thirty-two independent nulls would produce, and two of the three are positive, the wrong sign for a provision that lowers claims.¹⁷

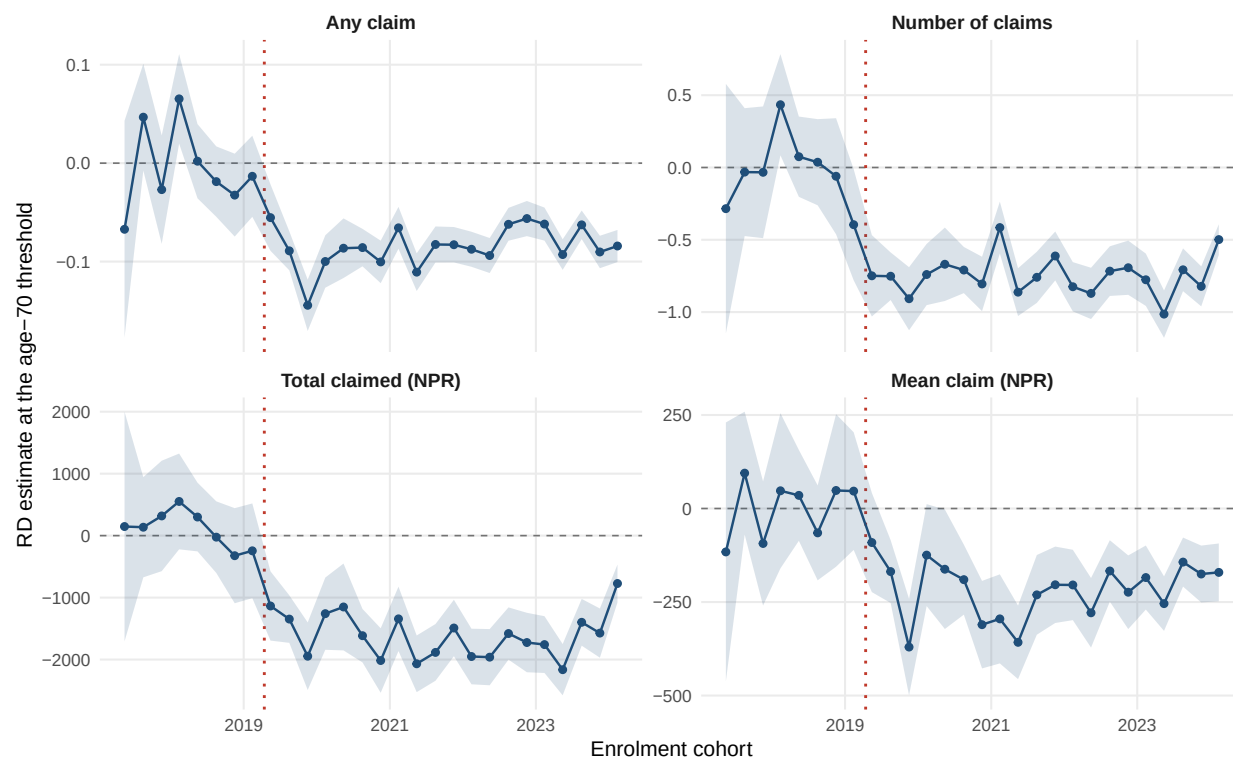
The discontinuity appears in the first treated quarter and never goes away. Enrollees starting in May 2019, the first cohort to face a zero premium at 70, already claim substantially less above the cutoff than below. Every one of the nineteen post-policy quarters is negative and significant at the 1 percent level, and the other three margins switch on at the same date. The coverage response and the claims response share the policy's timing.

The figure holds the window fixed at 1,095 days across every quarter and every margin, wider than the MSE-optimal bandwidth the main specification uses for any outcome. A common window is what makes the series comparable, because a bandwidth chosen quarter by quarter would make each point a different local estimand over a different span of ages. It is also what gives the placebo its power, since it lowers the average pre-policy standard error on total claimed. Because the

17. The third is the number-of-claims estimate in February 2019, the last quarter before the waiver, and it is smaller in magnitude than the estimate that arrives in the very next quarter and holds thereafter, with spending in that quarter flat. I have not been able to date the waiver's announcement as distinct from its implementation, so whether that single point is anticipation or noise the design cannot say.

window is wider, the quarterly estimates are larger in magnitude than the estimates of Table 5. The figure is evidence on when the discontinuity appears rather than on how large it is.

Figure 7 — The discontinuity appears only after the 2019 policy onset



Note: RD estimate at the age-70 threshold by enrolment quarter, with 95 percent confidence intervals. Local linear with separate slopes on each side, uniform kernel, no covariates, clustered on the household, estimated within a fixed window of $h = 1,095$ days for every quarter and every panel. The dotted line marks April 2019, when the free-at-70 provision took effect; the first cohort to face it starts in May 2019. Enrolment cohorts run from the 2017 cycle to February 2024. The final point is the smallest of the treated estimates, and that cohort's insurance year is only partly covered by the reimbursement records, which compresses claims on both sides of the cutoff. The estimates and standard errors are reported in Table B1.

6.2. Specification Robustness

The estimate survives the choices that govern a local-linear design. Table D1 in Appendix D.1 re-estimates all four claims outcomes across a bandwidth grid from 180 to 1,095 days, a donut that drops the quarter below the cutoff, a symmetric donut that drops that quarter on both sides, a triangular kernel, and a quadratic polynomial. Panel E is the main specification itself, at the bandwidth the MSE-optimal selector of Calonico, Cattaneo, and Titiunik (2014) returns for each outcome. Of the 96 cells, 92 are negative and 83 are significant at 5 percent.

The donut variants move the estimate away from zero rather than towards it. Dropping the quarter immediately below the cutoff, which removes any short-run timing of enrolment around the threshold, gives a larger discontinuity on both total claimed and any claim than the same window with that quarter retained, and dropping it on both sides does the same. Manipulation at the threshold would show up as an estimate that shrinks when that quarter is dropped. It grows instead, so the spells nearest the cutoff attenuate the discontinuity rather than create it.

Above the MSE-optimal window, magnitudes rise with the bandwidth. Wider windows fold in more of the age gradient in claims, so the larger estimates carry more specification bias rather than more information. A triangular kernel, which weights the observations nearest the cutoff most heavily, leaves the discontinuity negative and significant at the same window.

Thirteen cells are insignificant at 5 percent, and eleven of them are donut specifications at the two narrowest windows, where removing 91 days on one or both sides leaves too little of the window to identify a slope. The symmetric donut at $h = 180$ splits the sample by a 182-day gap, and its standard error on total claimed is six times the baseline. All four positive cells sit in that same corner. The remaining two insignificant cells are the quadratic at $h = 485$, where the second-order term is fitted over a window narrow enough that the estimate is imprecise; at the wider windows where a quadratic is identified it is negative and significant. No specification with the power to detect the effect reverses its sign.

6.3. Manipulation and Strategic Timing

The running variable is date of birth, recorded in administrative registration records and crossed by aging rather than by choice. It is anchored to the citizenship certificate or national identity document that NHIP enrolment requires, documents issued long before the waiver is in prospect, so manipulation here would mean falsifying an identity record rather than timing a decision. Misreporting in the registry cannot be ruled out a priori, which is what the density test addresses.

The standard manipulation concern is bunching of the running variable at the cutoff, and here the object to test is the density of the eligible population rather than of enrollees. An enrollee-level density test is the wrong instrument in this design, because enrolment responds to the price: the jump in enrollee density at 70 is the response the paper studies. Testing the census

population density instead, and correcting for age heaping, the log-density jump at 70 is -0.046 (SE 0.145), smaller than the same statistic at placebo ages on either side. Appendix A.3 reports both diagnostics. A third check comes from the estimate itself, since Section 6.2 shows that dropping the quarter on either side of the cutoff, where a falsified date of birth would place an enrollee, makes the discontinuity larger rather than smaller.

A second version of the concern is behavioural rather than mechanical. If enrollees let coverage lapse and timed re-entry to the waiver, the pool just above the cutoff would be selected on anticipated need. The enrolment-gap distribution shows the opposite. Gaps at the ages bracketing 70 are no longer than elsewhere, and enrollees whose gap spans their seventieth birthday claimed less than half as much before 70 as everyone else, where strategic timing predicts more. Appendix D.3 reports the distributions.

7 — Conclusion

National health insurance in low- and middle-income countries runs on enrolment that is voluntary in law or in practice, so the premium is what decides who is insured (WHO and Bank 2023; Banerjee et al. 2021). Nepal sets that price to zero at age 70, and the discontinuity shows what the price had been doing. Coverage nearly doubles at the threshold, and claims per enrollee fall across every margin. Enrolment responds to the price, so the contrast compares two pools rather than one pool before and after, and what falls is the cost of the pool the programme carries rather than the care a fixed population uses. The enrollees the waiver draws in cost NPR 3,358 each in paid claims against NPR 5,269 for the pool that was already paying, a ratio of 0.637 whose bootstrap interval, $[0.563, 0.729]$, lies wholly below one. The premium had been screening in the costlier, which is adverse selection into the paid programme below the threshold. The direction matches the one experimental estimate from a comparable scheme, where time-limited subsidies in Indonesia's programme also drew in enrollees of lower expected cost (Banerjee et al. 2021).

The waiver extends coverage into the places where the gap was. The enrollees it reaches live 0.54 km farther from their assigned first point of contact, have 0.74 fewer empanelled facilities within 10 km of their ward, and live in local units poorer by 1.7 percentage points of the poverty headcount. The coverage it adds goes to people who claim less once insured.

The threshold is informative because of where it sits relative to the programme's own defences. NHIP bundles coverage at the household level, which suppresses selection within the household (Fischer, Frölich, and Landmann 2023), and confines enrolment to fixed quarterly windows. Both are in force below 70, and the pool insured under them is adversely selected even so. The waiver does not let selection in. It reveals what the premium had already been doing, because the enrollees a zero price draws in are the ones the price had been keeping out. In Indonesia the binding constraint was administrative capacity, with more than half of attempted enrolments failing to complete (Banerjee et al. 2021); here the rules operate as written and the seam is in the price. A programme can be well run and well defended and still be priced in a way that decides who carries its risk.

What the waiver costs is easier to state than what it is worth. For every enrollee it adds, about one was insured at the premium already and now pays nothing, so roughly half of the budget buys no new coverage while the rest insures enrollees who cost NPR 3,358 a year. Valuing that against what those enrollees gain would require a price for coverage below the threshold, and the household schedule supplies no single one (Section 5.4). What the discontinuity settles is the cost side, and it settles it without a price.

The composition of the pool improves as coverage grows, since the enrollees a lower price draws in cost less than those already insured. What tightens is the financing. A waiver lowers the cost of insuring each enrollee while raising the total the programme must find, and it does so as premium revenue stops growing, so the binding constraint moves from the composition of the pool to the budget that carries it. Nepal has reached that point. Arrears owed to empanelled hospitals passed NPR 11 billion by late 2025 and more than fifty hospitals had suspended acceptance of insurance cards by early 2026, while the Board responded by capping outpatient coverage rather than by widening the base (The Kathmandu Post 2026, 2025). A programme reaches the members a price will never reach only by not charging them, and it can carry them only if the pool it subsidises also holds members who would have paid. Compulsion supplies those members where it can be enforced, as it can on formal employment, which makes it the one instrument that moves composition and revenue in the same direction. This study does not test that route, though it does establish what the alternatives leave untouched, since restricting when the willing may enrol, or the unit in which they must enrol, changes neither the composition of the pool nor the revenue

behind it.

Three limitations bound the reading. The estimates are local, to age 70 and to the enrollees a zero price draws in. They say who enrolls and what that pool costs, so any moral-hazard response in the marginal enrollees' use of care is left unmeasured. And the size of the gap moves with the bandwidth, from 0.554 to 0.728 across windows of one to three years, while rupee costs are understated on both sides as reimbursement records accrue; the ordering that defines adverse selection survives both, which is why I report the ratio. Whether that lower cost reflects better health or lighter use of care, and whether the waiver improves the outcomes insurance is meant to protect, the claims register cannot settle, since it observes only reimbursed use among the insured. None of them disturbs what the age-70 discontinuity makes precise, which is what a government buys, and for whom, when it makes insurance free.

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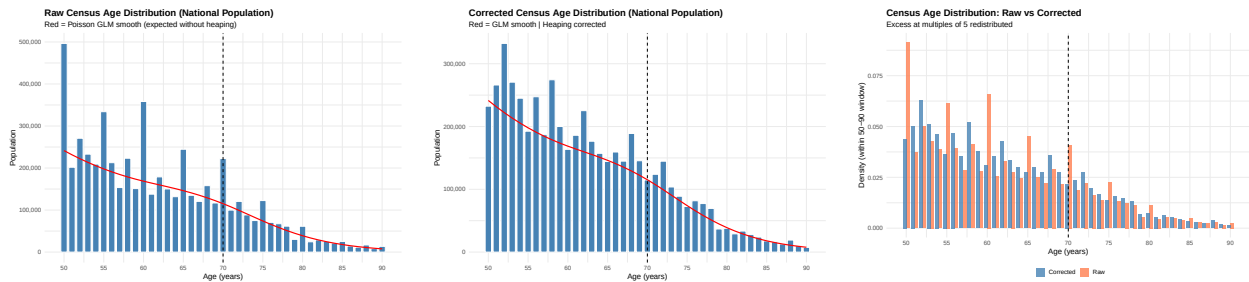
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Appendix A — Design Validity

A.1. Census Age-Heaping Correction

The covariate-balance and density tests of Section 3 and Appendix A.3 are run on the 2021 Nepal Census, in which respondents disproportionately report ages at multiples of five. Age 70 is both the policy threshold and a heaping spike, so the raw counts overstate the population at 70 and understate it at neighbouring ages. I correct for this by stochastic redistribution (Heitjan and Rubin 1990). A quasi-Poisson model fitted to single-year counts at non-heaped ages predicts a smooth density through the multiples of five; the excess mass at each heaped age, the gap between the observed count and the smooth prediction, is reassigned to neighbouring ages in proportion to the smooth density. Figure A1 shows the raw counts, the corrected counts, and the two overlaid. The procedure is repeated $M = 50$ times with independent reassignments, and every estimate that uses the corrected counts is computed within each draw and combined across draws by Rubin’s rules (Rubin 1987), so the reported standard errors carry both sampling uncertainty and the uncertainty introduced by the correction.

Figure A1 — Age heaping in the 2021 census and its correction



Note: Single-year population counts from the 2021 Nepal Census. Left: raw counts, with spikes at multiples of five. Centre: counts after stochastic redistribution. Right: the two overlaid. The age-70 spike is comparable to spikes at non-policy multiples of five.

The balance result is robust to the bandwidth of the test and to the choice of cutoff. The tables below re-run the covariate balance under alternative bandwidths and at placebo cutoffs of 63 and other non-policy ages; the age-70 nulls of Section 3.2 hold throughout, and the placebo ages return discontinuities of comparable or larger magnitude.

Table A1 — Covariate Balance at Age 70: Sensitivity to Kernel and Polynomial Order

	Mean	Linear ($p = 1$)		Quadratic ($p = 2$)	
	Pre-70	Uniform	Triang.	Uniform	Triang.
Female	0.509	0.013* (0.007)	0.012 (0.009)	0.011 (0.015)	0.034* (0.020)
High Caste	0.478	0.000 (0.007)	−0.007 (0.009)	−0.016 (0.015)	0.008 (0.021)
Hindu	0.810	0.010* (0.006)	0.008 (0.007)	0.006 (0.011)	0.036** (0.016)
Ever married	0.988	−0.000 (0.002)	0.001 (0.002)	0.002 (0.003)	0.003 (0.005)
Has disability	0.064	0.000 (0.003)	−0.003 (0.004)	−0.008 (0.007)	−0.009 (0.010)
Household size	5.204	0.096 (0.113)	0.168 (0.126)	0.243 (0.208)	0.491 (0.372)
Economically inactive (aged)	0.354	0.009 (0.007)	−0.000 (0.009)	−0.012 (0.014)	0.000 (0.020)
Literate	0.339	−0.013* (0.007)	−0.025*** (0.009)	−0.036*** (0.014)	−0.112*** (0.019)
Wealth quintile	2.875	−0.039* (0.022)	−0.055* (0.029)	−0.076* (0.044)	−0.267*** (0.061)
Bandwidth (years)	4				
Observations	138,314				

Notes: Each cell reports the coefficient on the treatment indicator from a local regression estimated by OLS (feols) with standard errors clustered at the household level. Bandwidth is 4 years on each side of age 70. Columns vary the polynomial order and kernel weighting. Estimates are pooled across 50 Monte Carlo draws of stochastic age redistribution using Rubin’s combination rules; standard errors account for both sampling and imputation uncertainty. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2 — Placebo Balance Test at Age 69 (No Policy Change)

	Mean	Linear ($p = 1$)		Quadratic ($p = 2$)	
	Pre-69	Uniform	Triang.	Uniform	Triang.
Female	0.510	0.003 (0.007)	−0.001 (0.008)	−0.008 (0.013)	−0.024 (0.020)
High Caste	0.479	0.006 (0.007)	0.001 (0.008)	−0.010 (0.013)	−0.031 (0.020)
Hindu	0.813	0.011** (0.005)	−0.003 (0.006)	−0.023** (0.010)	−0.037** (0.015)
Ever married	0.988	0.001 (0.001)	−0.001 (0.002)	−0.004 (0.003)	−0.002 (0.005)
Has disability	0.061	0.001 (0.003)	0.004 (0.004)	0.009 (0.006)	0.001 (0.009)
Household size	5.313	0.165* (0.090)	−0.091 (0.143)	−0.479 (0.335)	−0.281 (0.443)
Economically inactive (aged)	0.328	0.042*** (0.006)	0.013* (0.008)	−0.030** (0.012)	−0.014 (0.019)
Literate	0.339	−0.036*** (0.006)	0.019** (0.008)	0.106*** (0.013)	0.089*** (0.019)
Wealth quintile	2.867	−0.077*** (0.020)	0.054** (0.023)	0.248*** (0.039)	0.242*** (0.060)
Bandwidth (years)	4				
Observations	144,613				

Notes: Each cell reports the coefficient on the treatment indicator from a local regression estimated by OLS (feols) with standard errors clustered at the household level. Bandwidth is 4 years on each side of age 69. Columns vary the polynomial order and kernel weighting. Estimates are pooled across 50 Monte Carlo draws of stochastic age redistribution using Rubin’s combination rules; standard errors account for both sampling and imputation uncertainty. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A3 — Placebo Balance Test at Age 63 (No Policy Change)

	Mean	Linear ($p = 1$)		Quadratic ($p = 2$)	
	Pre-63	Uniform	Triang.	Uniform	Triang.
Female	0.508	0.009 (0.006)	0.003 (0.008)	−0.005 (0.013)	−0.052*** (0.019)
High Caste	0.469	0.006 (0.006)	−0.002 (0.007)	−0.010 (0.012)	−0.050*** (0.019)
Hindu	0.812	−0.000 (0.005)	−0.005 (0.006)	−0.009 (0.010)	−0.050*** (0.015)
Ever married	0.986	−0.002 (0.001)	−0.002 (0.002)	−0.002 (0.003)	−0.007 (0.004)
Has disability	0.051	0.001 (0.003)	−0.001 (0.003)	−0.003 (0.005)	−0.002 (0.008)
Household size	5.307	−0.072 (0.105)	−0.086 (0.127)	−0.041 (0.205)	−0.473 (0.364)
Economically inactive (aged)	0.133	−0.014*** (0.004)	0.031*** (0.006)	0.103*** (0.009)	−0.023 (0.014)
Literate	0.400	−0.002 (0.006)	−0.017** (0.007)	−0.049*** (0.012)	0.146*** (0.019)
Wealth quintile	2.925	0.018 (0.018)	−0.010 (0.023)	−0.069* (0.037)	0.395*** (0.057)
Bandwidth (years)	4				
Observations	174,496				

Notes: Each cell reports the coefficient on the treatment indicator from a local regression estimated by OLS (feols) with standard errors clustered at the household level. Bandwidth is 4 years on each side of age 63. Columns vary the polynomial order and kernel weighting. Estimates are pooled across 50 Monte Carlo draws of stochastic age redistribution using Rubin's combination rules; standard errors account for both sampling and imputation uncertainty. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.2. Covariate Balance: Age Profiles

Figure A2 displays the age profile of each census covariate, with separate linear fits on each side of the threshold. The visual continuity at age 70 matches the small, jointly insignificant point estimates of Section 3.2.

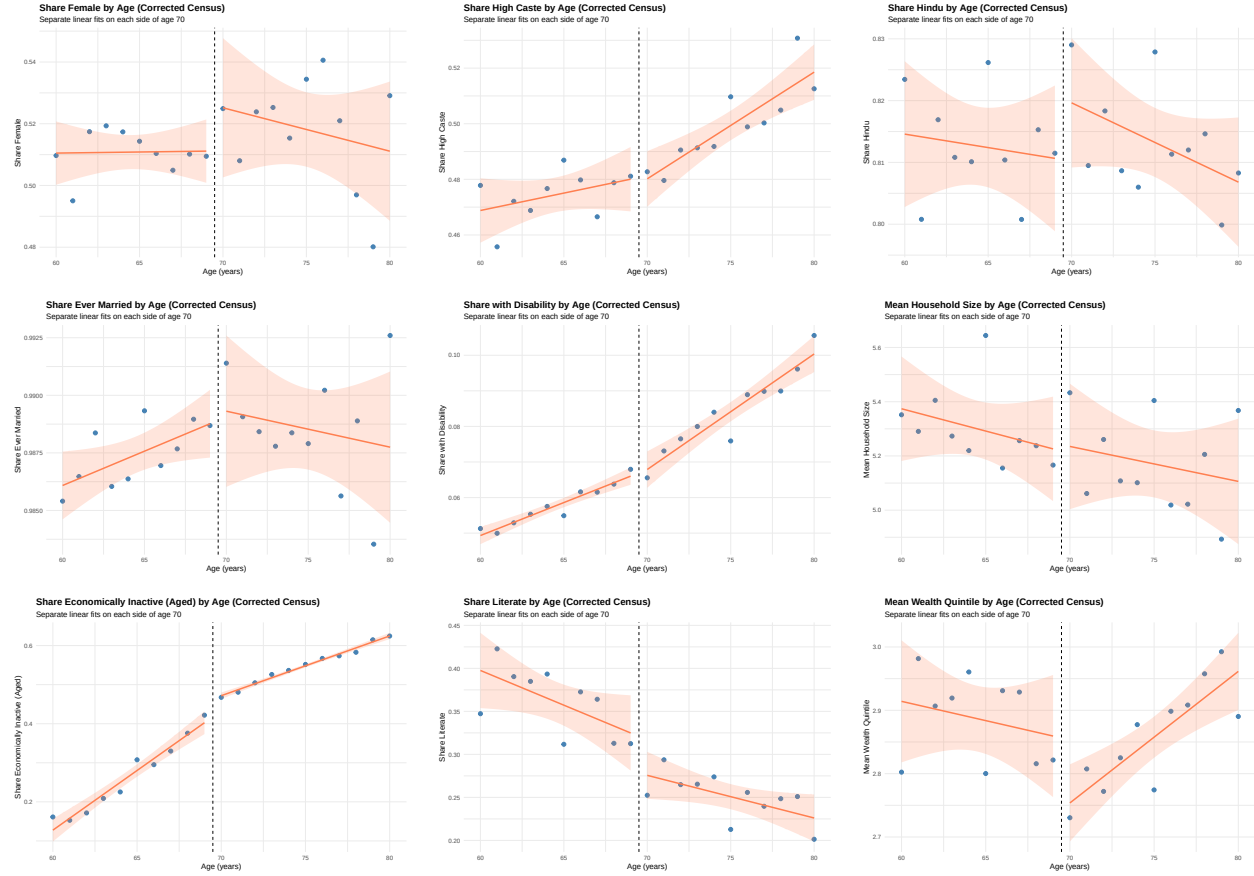


Figure A2 — Covariate age profiles around the age-70 threshold (2021 Nepal Census). The panels are the nine covariates of Table 3, in the order they appear there. Each shows the weighted mean by age with separate linear fits on each side of the cutoff; ages corrected for heaping (Appendix A.1).

A.3. Density of the Eligible Population

The standard manipulation concern in a regression discontinuity design is bunching of the running variable at the cutoff. Here the relevant smoothness object is the density of the *eligible population*, everyone aging through 70, not the density of enrollees: an enrollee-level McCrary-style test is the wrong object in this design because enrolment itself responds to the price change, so the jump

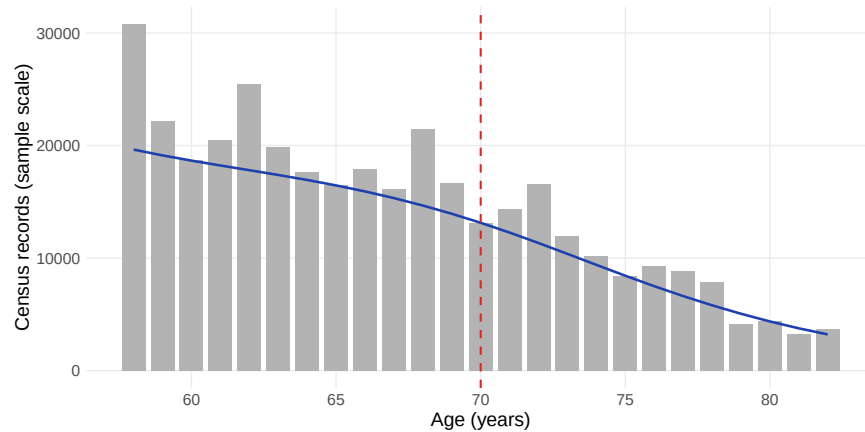
in enrollee density at 70 is the enrolment response the paper studies, not manipulation of the running variable. The running variable itself is date of birth in administrative registration records, crossed by aging rather than by choice; the recorded date of birth is anchored to the citizenship certificate or national identity document that NHIP enrolment requires, though misreporting in the registry cannot be ruled out a priori. The census density test addresses manipulation at the level of the eligible population; evidence on enrollee-side irregularities sits with the placebo checks of Section 3.3 and the inference diagnostics of Appendix D.2.

I therefore test the smoothness of the census population density at 70, again correcting for age heaping. Table A4 reports both diagnostics. Panel A compares observed single-year counts to a smooth fit excluding multiples of five: the raw heap at age 70 is 1.94 times the smooth count, within the 1.42–2.18 range spanned by non-policy multiples of five, and smaller than the heap at age 60 (2.18), where no insurance policy applies. Round-age reporting, not strategic misreporting around the threshold, accounts for the spike. Panel B fits a quasi-Poisson local-linear model to the heaping-corrected counts over ages 62–78: the log-density jump at 70 is -0.046 (SE 0.145, $p = 0.753$), and the same statistic at non-policy placebo ages is larger in magnitude on both sides (0.296 at 67, -0.226 at 73). The age-70 estimate is small relative to the ordinary fluctuation of the corrected density at ages with no policy content. Figure A3 plots the corrected density.

Table A4 – Population density at the age-70 threshold

<i>Panel A: Raw census — excess mass at heaping points (observed/expected)</i>							
Age	55	60	65	70	75	80	85
Excess ratio	1.74	2.18	1.69	1.94	1.68	1.62	1.42
<i>Panel B: Corrected census — log-density jump (Rubin-pooled over 50 draws)</i>							
	At 70	At 67 (placebo)	At 73 (placebo)				
Jump	-0.0456	0.2957	-0.2256				
SE	(0.1447)	(0.1514)	(0.1468)				

Notes: Panel A compares observed single-year age counts to a quasi-Poisson spline fit estimated excluding multiples of five: the excess at 70 is in line with the excess at 65 and 75, which carry no policy discontinuity, indicating round-age reporting rather than manipulation. Panel B fits a quasi-Poisson local linear model to heaping-corrected single-year counts (ages 62–78) and reports the log jump at the stated cutoff, combined across 50 redistribution draws with Rubin’s rules. The enrollee-level density is not tested: its jump at 70 is the enrollment response to the price change, not manipulation of date of birth.

Figure A3 – Distribution of Eligible Population

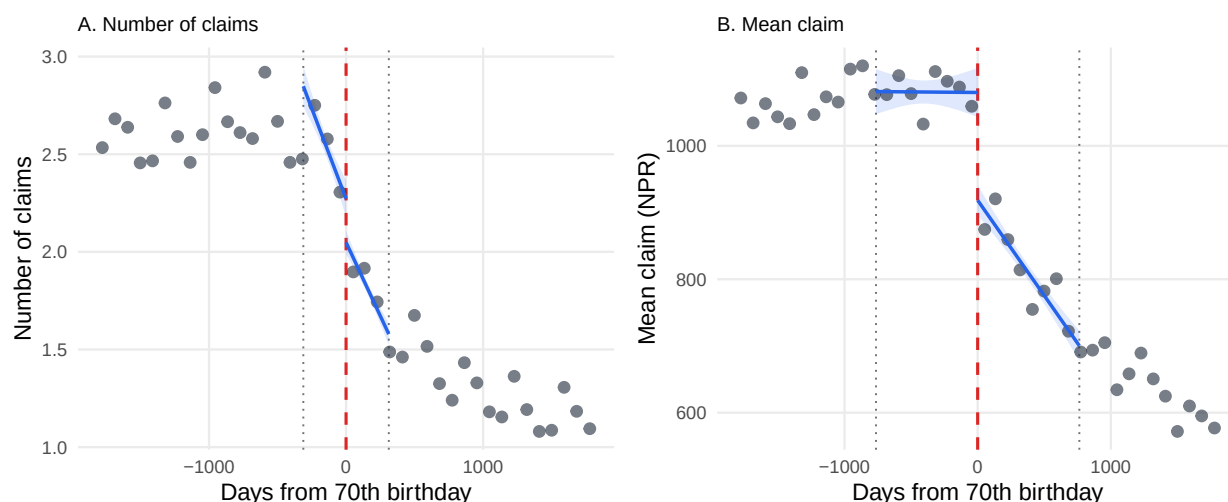
Note: Single-year age counts of the eligible population from the 2021 Nepal Census, corrected for age heaping (Appendix A.1). The dashed vertical line marks age 70. Table A4 reports the formal test of the log-density jump at the threshold and the placebo jumps at ages 67 and 73.

Appendix B — Additional Claims Results

B.1. The Remaining Claims Margins

Figure B1 plots the two claims margins the main text leaves out, the number of claims and the mean claim, where Figure 4 shows any claim and total claimed. Both step down at the threshold in the same way, which is the graphical counterpart of Table 5.

Figure B1 — The discontinuity on the remaining two claims margins



Note: Drawn as Figure 4 is drawn. Binned means in quarterly bins of age at the spell start date, shown over twenty quarters on each side of the seventieth birthday, with local linear fits estimated on each side of the cutoff on the spell-level data inside each panel's own bandwidth, a uniform kernel and no covariates or fixed effects. The bandwidth is 312 days for the number of claims and 764 for the mean claim. The gap drawn at the cutoff is the coefficient reported in Table 5.

B.2. Pre-Policy Event Study Estimates

Table B1 reports the estimates plotted in Figure 7, quarter by quarter and for all four claims margins, so the pre-policy nulls can be read as numbers rather than measured off a confidence ribbon.

B.3. Timing of the First Claim

Table B2 sets out the three timing outcomes reported in Section 4.2. The running variable, the kernel, the clustering and the inference are those of Table 5, and each column takes its own

Table B1 – RD estimates by enrolment quarter

Enrolment quarter	Any claim (1)	N claims (2)	Total claimed (NPR) (3)	Mean claim (NPR) (4)
<i>Panel A: Before the waiver</i>				
May 2017	-0.067 (0.056)	-0.285 (0.440)	146.9 (945.9)	-116.1 (176.6)
Aug 2017	0.047 (0.028)	-0.032 (0.226)	135.7 (412.7)	94.5 (83.9)
Nov 2017	-0.027 (0.028)	-0.033 (0.232)	317.8 (454.9)	-93.5 (84.7)
Feb 2018	0.065 (0.023)	0.434 (0.179)	551.7 (394.4)	47.4 (105.8)
May 2018	0.002 (0.019)	0.075 (0.142)	300.2 (283.1)	35.1 (62.0)
Aug 2018	-0.019 (0.018)	0.036 (0.152)	-24.8 (295.0)	-65.1 (64.7)
Nov 2018	-0.032 (0.021)	-0.061 (0.205)	-324.0 (391.6)	48.2 (104.0)
Feb 2019	-0.013 (0.021)	-0.396 (0.196)	-245.0 (390.1)	46.5 (80.3)
<i>Panel B: After the waiver</i>				
May 2019	-0.055 (0.017)	-0.749 (0.144)	-1,135.5 (285.2)	-90.9 (67.7)
Aug 2019	-0.089 (0.010)	-0.752 (0.084)	-1,346.9 (195.7)	-168.6 (42.8)
Nov 2019	-0.144 (0.013)	-0.908 (0.111)	-1,946.0 (278.5)	-370.4 (65.8)
Feb 2020	-0.100 (0.014)	-0.740 (0.108)	-1,259.5 (297.5)	-124.6 (69.5)
May 2020	-0.086 (0.015)	-0.669 (0.130)	-1,151.5 (357.7)	-162.3 (81.5)
Aug 2020	-0.086 (0.010)	-0.709 (0.082)	-1,615.0 (220.2)	-190.2 (47.7)
Nov 2020	-0.100 (0.011)	-0.805 (0.096)	-2,015.6 (265.6)	-310.5 (59.7)
Feb 2021	-0.066 (0.011)	-0.416 (0.092)	-1,343.4 (265.3)	-295.1 (60.8)
May 2021	-0.111 (0.010)	-0.862 (0.084)	-2,068.5 (233.3)	-357.7 (50.2)
Aug 2021	-0.083 (0.009)	-0.759 (0.090)	-1,884.7 (233.6)	-230.8 (54.4)
Nov 2021	-0.083 (0.009)	-0.612 (0.086)	-1,490.9 (232.6)	-203.9 (52.1)
Feb 2022	-0.087 (0.009)	-0.825 (0.087)	-1,950.9 (228.9)	-204.4 (48.0)
May 2022	-0.094 (0.009)	-0.871 (0.091)	-1,961.7 (231.6)	-278.8 (47.4)
Aug 2022	-0.062 (0.009)	-0.717 (0.088)	-1,580.2 (215.2)	-167.0 (41.9)
Nov 2022	-0.056 (0.009)	-0.693 (0.096)	-1,725.3 (244.9)	-223.8 (50.2)
Feb 2023	-0.062 (0.009)	-0.777 (0.092)	-1,758.6 (233.9)	-184.6 (43.7)
May 2023	-0.093 (0.008)	-1.014 (0.084)	-2,163.5 (211.3)	-254.3 (37.2)
Aug 2023	-0.063 (0.007)	-0.707 (0.075)	-1,398.9 (192.5)	-143.2 (33.4)
Nov 2023	-0.090 (0.008)	-0.822 (0.071)	-1,573.1 (202.6)	-175.1 (38.8)
Feb 2024	-0.084 (0.008)	-0.499 (0.054)	-772.5 (156.2)	-170.7 (39.4)

Notes: The estimates plotted in Figure 7. Each cell is a local linear RD at the age-70 threshold estimated on that enrolment quarter alone, with separate slopes on each side, a uniform kernel, no covariates, and a window fixed at $h = 1,095$ days throughout. Standard errors clustered on the household in parentheses. The free-at-70 provision took effect in April 2019, so the first quarter of Panel B is the first cohort to face a zero premium at 70.

mean-squared-error-optimal bandwidth. Those bandwidths run from 379 days to 803, so the three columns rest on different windows and on samples ranging from 86,908 spells to 114,357.

The two indicator columns are defined for every spell and are zero for an enrollee who never claims, so each combines whether a claim happens at all with how soon it happens. Days to the first claim is defined only among enrollees who do claim, and its below-cutoff mean of 67.7 days is therefore an average over a pool whose composition changes at the threshold.

Table B2 — RD estimates at the age-70 threshold, timing of the first claim

	Days to first claim (days) (1)	Claim within 30 days (0/1) (2)	Claim within 90 days (0/1) (3)
RD estimate	7.36*** (1.05)	−0.0492*** (0.0050)	−0.0574*** (0.0064)
Robust <i>p</i> -value	<0.001	<0.001	<0.001
Mean below cutoff	67.7	0.225	0.369
% change	10.9%	−21.9%	−15.5%
Bandwidth (days)	803	440	379
Observations	86,908	114,357	96,149

Notes: Each observation is an enrolment spell. The running variable, specification, bandwidth rule, clustering and inference are those of Table 5. Column 1 is defined only for spells that claim, so it conditions on a post-treatment outcome and is descriptive. Columns 2 and 3 are defined for every spell, non-claimers included, and are zero for a spell that never claims. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.4. Distance to Facilities

Table B3 reports the total-claimed discontinuity estimated separately within distance bands from the enrollee’s ward centroid to the registered first point of contact. Every band is fitted on the same 485-day window as the headline estimate, so the bands differ only in distance and not in the window each is estimated on.

The estimates are NPR −734.7 (SE 186.7) within 5 km, NPR −1,114.0 (SE 254.8) between 5 and 10 km, NPR −644.1 (SE 328.9) between 10 and 15 km, and NPR −933.9 (SE 319.4) beyond 15 km, against a pooled NPR −918.0 (SE 127.3) on the 125,859 spells with both a geocoded ward

and a geocoded facility. The pooled figure reproduces the headline estimate of Table 5, the small residual reflecting the 1.2 percent of spells without geocodes.

No band differs significantly from any other. The nearest and most remote bands differ by NPR 199 against a standard error of roughly 370, and the point estimates are not monotone in distance, since the largest drop falls in the 5 to 10 km band and the smallest between 10 and 15 km. A decline driven by the marginal pool's remoteness would deepen steadily with distance, and these estimates show no such gradient.

Table B3 — RD estimates at the age-70 threshold, total claimed by distance to the first point of contact

	0–5 km (1)	5–10 km (2)	10–15 km (3)	15 km and over (4)	Pooled (5)
RD estimate	-734.7*** (186.7)	-1,114.0*** (254.8)	-644.1* (328.9)	-933.9*** (319.4)	-918.0*** (127.3)
Mean below cutoff	6,103	4,734	4,399	3,477	5,290
% change	-12.0%	-23.5%	-14.6%	-26.9%	-17.4%
Bandwidth (days)	485	485	485	485	485
Observations	62,755	31,540	15,351	16,213	125,859

Notes: Each observation is an enrolment spell. The outcome is the total amount claimed in NPR over the spell, zero for a spell with no claims. Distance is the Haversine distance in kilometres from the enrollee's ward centroid to the registered first point of contact. Each band is fitted separately, and all are fitted on the same bandwidth, the mean-squared-error-optimal choice for total claimed in Table 5, held fixed so that the bands differ only in distance. Column 5 pools the four bands. Standard errors in parentheses are clustered on the household. Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix C — Selection and Cost

C.1. Bandwidth Sensitivity of the Cost Estimates

Table C1 recomputes the two average costs at four bandwidths and recovers $\overline{MC}$ from the identity at each. The coverage shares are held at their Table 6 values throughout, because they come from the point-in-time insured stock rather than from the claims regression and so do not depend on this bandwidth.

AC_{below} is stable across the four rows, moving only between 5,117 and 5,279, while $\overline{MC}$ falls from 3,726 to 2,924 as the window widens. The identity writes $\overline{MC}$ as AC_{below} plus $Q_{\text{above}}/\Delta Q$ times the gap between the two average costs, and that multiplier is close to two, so a window that estimates a larger gap moves $\overline{MC}$ by roughly twice as much. The fall across the rows therefore reflects a larger estimated discontinuity in average cost at wider windows rather than any change in the pool that was already paying.

The ordering $\overline{MC} < AC_{\text{below}}$ holds at every bandwidth, so the qualitative finding does not turn on the choice. The magnitude does. The ratio runs from 0.554 to 0.728 across the four rows, about the width of the bootstrap interval at the main bandwidth, $[0.563, 0.729]$. Bandwidth choice and sampling uncertainty therefore move the ratio by comparable amounts, and neither carries it near one.

Table C1 — The ratio across bandwidths for the average costs

Specification	AC_{below}	$\overline{MC}$	$\overline{MC}/AC_{\text{below}}$
$h = 365$ days	5,117	3,726	0.728
$h = 485$ days (main)	5,269	3,358	0.637
$h = 730$ days	5,240	3,160	0.603
$h = 1095$ days	5,279	2,924	0.554

Notes: The bandwidth is varied for both average costs, and $\overline{MC}$ is recomputed from the identity at each one. Coverage shares are held at their Table 6 values, since they come from the point-in-time stock and do not depend on this bandwidth. Point estimates only. The ordering $\overline{MC} < AC_{\text{below}}$ holds at every bandwidth; the magnitude of the gap does not.

C.2. How Often the Benefit Ceiling Binds

Table C2 reports how far into the benefit ceiling enrolment spells reach on each side of the threshold. It is the evidence behind the contract-equivalence condition of Section 5.4: the ceiling changes unit at 70, from a household total to an individual entitlement, but it binds for a negligible share of spells either way, so the change in unit is not what moves realised claims.

Table C2 – How often does the coverage ceiling bind?

Group	P90	P99	Share $\geq$ 80k	Share $\geq$ 100k	N
Household-spells below 70 (shared 100k ceiling)	20,317	64,226	0.48%	0.09%	288,399
Individual spells 70+ (individual 100k ceiling)	8,225	43,214	0.17%	0.02%	822,009

Notes: Claimed amounts within the analysis sample (spells starting May 2023–May 2024, ages 60–80). Below 70 the NPR 100,000 ceiling is shared at the household level, so totals are summed over household-spell; at and above 70 the ceiling is individual.

Appendix D – Robustness and Inference

D.1. Robustness of the Main Estimates

Table D1 reports the full set of specification checks summarised in Section 6.2: bandwidth sensitivity, donut and symmetric-donut variants, a triangular kernel, quadratic polynomials at wider bandwidths, and the Calonico, Cattaneo, and Titiunik (2014) MSE-optimal data-driven bandwidth. No specification overturns the sign or significance of the main estimates.

D.2. Randomisation and Inference

The placebo logic extends to a randomisation distribution denser than the whole-year grid of Figure 2. I re-estimate the discontinuity at 50 placebo cutoffs spaced a quarter of a year apart across ages 62–68 and 72–78, each fitted only on spells lying on one side of age 70 so that no placebo window straddles the true threshold. Treating these 50 estimates as the distribution of discontinuities that arise where no insurance policy operates, the observed age-70 estimate exceeds every one of the 50 placebo estimates in magnitude for both any claim and total claimed; the implied permutation p-value is $1/51 \approx 0.02$, the smallest attainable with a grid of this size. Figure D1 plots the observed estimate against the placebo distribution; no placebo discontinuity approaches it in magnitude.

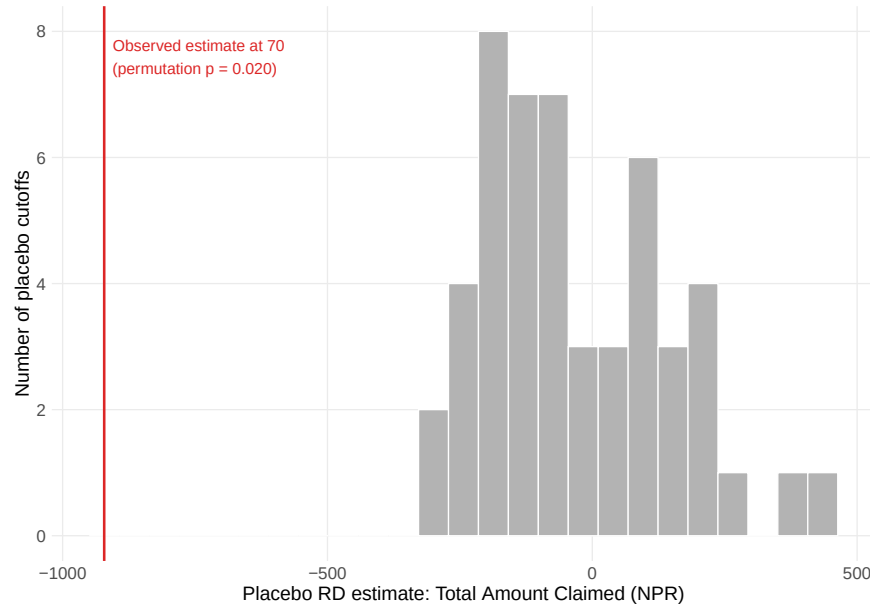
Inference is also robust to the choice of clustering. The baseline clusters by household, the level at which enrolment decisions and premiums are set. The running variable takes discrete

Table D1 – Robustness of the RD estimates to alternative specifications

	Any Claim (1)	N Claims (2)	Total Claimed (NPR) (3)	Mean Claim (NPR) (4)
<i>Panel A: Bandwidth sensitivity (uniform kernel, local linear, no donut)</i>				
$h = 180$ days	-0.053 (0.010)	-0.260 (0.076)	-639.3 (210.2)	-177.5 (48.5)
$h = 312$ days	-0.052 (0.007)	-0.223 (0.057)	-480.0 (158.1)	-135.6 (35.9)
$h = 485$ days	-0.066 (0.006)	-0.471 (0.046)	-921.5 (126.9)	-157.3 (27.5)
$h = 764$ days	-0.072 (0.005)	-0.470 (0.037)	-930.7 (101.6)	-160.1 (21.8)
$h = 1095$ days	-0.086 (0.004)	-0.611 (0.031)	-1,135.0 (84.9)	-171.5 (17.9)
<i>Panel B: Donut (drop the quarter below the cutoff, $-91 \leq rv < 0$)</i>				
$h = 180$ days	-0.046 (0.032)	-0.113 (0.258)	-192.5 (697.0)	6.3 (123.8)
$h = 312$ days	-0.066 (0.013)	-0.393 (0.106)	-857.6 (287.2)	-120.4 (55.9)
$h = 485$ days	-0.083 (0.008)	-0.731 (0.067)	-1,460.8 (181.0)	-195.6 (34.3)
$h = 764$ days	-0.080 (0.006)	-0.580 (0.047)	-1,168.1 (126.2)	-174.7 (25.1)
$h = 1095$ days	-0.094 (0.004)	-0.702 (0.036)	-1,300.1 (98.2)	-179.4 (19.6)
<i>Panel B2: Symmetric donut (drop $rv \leq 91$ days)</i>				
$h = 180$ days	-0.007 (0.037)	0.467 (0.291)	981.2 (793.6)	148.0 (149.0)
$h = 312$ days	-0.048 (0.015)	-0.135 (0.116)	-208.0 (317.0)	-32.4 (64.4)
$h = 485$ days	-0.077 (0.009)	-0.679 (0.072)	-1,263.5 (195.4)	-145.6 (38.3)
$h = 764$ days	-0.079 (0.006)	-0.564 (0.049)	-1,088.7 (132.8)	-153.4 (27.0)
$h = 1095$ days	-0.096 (0.005)	-0.719 (0.038)	-1,303.0 (101.8)	-171.2 (20.7)
<i>Panel C: Triangular kernel (local linear)</i>				
$h = 180$ days	-0.057 (0.011)	-0.387 (0.083)	-961.6 (229.5)	-211.0 (55.9)
$h = 312$ days	-0.054 (0.008)	-0.265 (0.062)	-627.9 (172.5)	-161.9 (40.6)
$h = 485$ days	-0.058 (0.007)	-0.339 (0.050)	-692.7 (137.6)	-143.7 (31.5)
$h = 764$ days	-0.068 (0.005)	-0.447 (0.040)	-903.4 (110.9)	-159.3 (24.5)
$h = 1095$ days	-0.076 (0.004)	-0.528 (0.034)	-1,015.3 (92.9)	-162.5 (20.2)
<i>Panel D: Quadratic polynomial at the wider windows (uniform kernel)</i>				
$h = 485$ days	-0.046 (0.009)	-0.132 (0.069)	-342.2 (190.1)	-125.3 (43.8)
$h = 764$ days	-0.062 (0.007)	-0.412 (0.055)	-862.9 (150.6)	-158.1 (33.5)
$h = 1095$ days	-0.061 (0.006)	-0.402 (0.046)	-834.1 (126.3)	-148.5 (27.7)
<i>Panel E: Data-driven bandwidth (rdrobust, uniform, $p = 1$)</i>				
MSE-optimal	-0.059 [0.008]	-0.223 [0.060]	-921.5 [140.8]	-161.6 [23.6]
h (days)	342	312	485	764

Notes: The running variable is days from the seventieth birthday. Every cell is a local linear fit with separate slopes on each side, no covariates and no fixed effects, on the 2023–24 enrolment cycle. Panels A–D report feols standard errors clustered on the household in parentheses. Panel E reports the specification of Table 5: the MSE-optimal bandwidth of Calonico, Cattaneo, and Titiunik (2014), selected separately for each outcome and shown in the last row, with robust bias-corrected standard errors in brackets. The selected bandwidth differs by outcome, so no row of Panel A is the main specification for more than one column; Panel E is.

Figure D1 — The age-70 discontinuity lies outside the permutation distribution



Note: The histogram shows RD estimates for total claimed from 50 placebo cutoffs spaced a quarter of a year apart across ages 62–68 and 72–78, each estimated with the specification of Table 5, local linear in age in days with a uniform kernel, no covariates and no fixed effects, at the outcome’s own mean-squared-error-optimal bandwidth; the vertical line marks the observed age-70 estimate. The distribution for any claim is analogous. The observed estimate exceeds all 50 placebo estimates in magnitude for both outcomes, so the inclusive permutation p-value, $(1 + \#\{|\hat{\tau}_{\text{plac}}| \geq |\hat{\tau}_{\text{obs}}|\}) / (1 + 50) = 1/51 \approx 0.02$, is the smallest attainable with a grid of this size and is the value annotated on the figure.

values, so specification error common to all spells sharing a value of it is a residual concern (Lee and Card 2008), and Table D2 re-computes the standard errors clustering on that value instead. Measured in days the running variable is finely divided, between 623 and 1,527 distinct values depending on the column, so the few-cluster problem that attends coarser groupings does not arise here. The standard errors rise, from 0.007 to 0.009 for any claim, from 0.057 to 0.078 for the number of claims, from 126.9 to 167.1 for total claimed, and from 21.9 to 23.3 for the mean claim. Every estimate remains significant at the 1 percent level under the wider standard errors, the number of claims by the narrowest margin.

D.3. Strategic Enrolment Timing

If enrollees timed their enrolment to coincide with the free-premium threshold, letting coverage lapse until 70, the pool just above the cutoff would be selected on anticipated need, and the discontinuity would reflect strategic timing rather than price-driven selection. The enrolment-

Table D2 – Inference robustness: clustering by running variable

	Any claim (0/1) (1)	Number of claims (count) (2)	Total claimed (NPR) (3)	Mean claim (NPR) (4)
RD estimate	-0.059	-0.223	-921.5	-161.6
SE, clustered by household	(0.007)	(0.057)	(126.9)	(21.9)
SE, clustered by running variable	(0.009)	(0.078)	(167.1)	(23.3)
Bandwidth (days)	342	312	485	764
Running-variable cells	685	623	971	1,527
Spells	88,148	81,242	127,359	205,305

Notes: The specification is the one used in the main claims table: local linear in age in days with separate slopes on each side of the cutoff, a uniform kernel, no covariates and no fixed effects, each column at its own mean-squared-error-optimal bandwidth. The second row clusters on the household, as the main table does. The third clusters on the value of the running variable, which addresses specification error common to spells sharing a running-variable cell (Lee and Card 2008).

gap distribution gives no sign of it. The tables below report the distribution of gaps between consecutive spells by age and a comparison of pre-70 claims for enrollees whose gap spans their seventieth birthday against all others. Gaps at the ages bracketing 70 are no longer than elsewhere, and the gap-spanners record NPR 8,757 in pre-70 claims against NPR 19,354 for the rest: lower, not higher, than the comparison group, the opposite of what a strategic-timing story predicts.

Table D3 — Enrollment gap patterns by age at renewal (%)

Age	No gap	1 quarter	2–3 quarters	4+ quarters
60	75.5	11.0	5.7	7.8
61	75.4	11.1	5.7	7.8
62	75.6	11.0	5.7	7.6
63	76.0	10.8	5.6	7.6
64	75.9	10.9	5.6	7.6
65	76.1	10.8	5.5	7.6
66	75.9	11.0	5.6	7.4
67	76.2	10.8	5.6	7.4
68	76.6	10.7	5.3	7.4
69	76.9	10.7	5.3	7.1
70	78.8	9.5	5.1	6.7
71	91.5	3.7	2.0	2.8
72	94.6	2.2	1.3	1.9
73	95.9	1.7	1.0	1.5
74	96.5	1.4	0.8	1.3
75	96.9	1.3	0.7	1.1
76	97.0	1.2	0.7	1.1
77	97.1	1.2	0.6	1.1
78	97.2	1.1	0.6	1.1
79	97.2	1.1	0.6	1.1

Table D4 — Pre-70 claims by enrollment transition type

Transition type	N	Mean total claimed (NPR)	Mean no. of claims	% with any claim
<i>Panel A: By gap length</i>				
Continuous (no gap)	88,036	20,925	10.5	68.5
Short gap (< 1 quarter)	7,920	17,669	8.6	72.3
Medium gap (1–2 quarters)	7,973	14,529	7.1	67.8
Long gap (> 2 quarters)	25,344	7,750	3.7	54.1
<i>Panel B: Gap spans 70th birthday</i>				
No	109,682	19,354	—	67.6
Yes	19,591	8,757	—	56.4